Media Wall & Dining Wall: Build Guide

Version 13 · Oct 18, 2026 · for Emma and Dave · sizes in cm; the full plan has the drawings and the fit proof



This guide assumes you've hung shelves and put flat-pack together, but haven't cut a joint before. Every step says what to do, what to check before moving on, and roughly how long it takes the first time. Where a step is a job for a professional, it says so. Sizes are finished sizes in centimetres.

Before you start: five rules

- **Check every wall before you drill.** Run the cable and pipe detector over the spot, and never drill straight above, below or beside a socket or switch: cables run vertically and horizontally from them.
- **Measure twice, mark once, cut once.** Walls are rarely square: measure the alcove at the floor, 76, 134, 155, 196 and the ceiling, and cut to the smallest.
- **Two people for anything over 15 kg**: the uprights, the worktop, the table top, the TV.
- **Mains wiring is the electrician's.** Swapping the socket, the new socket behind the sofa and the dimmers are all theirs. Adding to existing circuits like this isn't normally notifiable under Part P, but a registered electrician certifies it either way.
- **Dry-fit everything** (assemble without glue or screws) before the final fix.

Tools

You don't need a workshop. Hire the two big tools for a weekend (a track saw and an SDS drill), or ask the timber merchant to cut the oak and ply to the cut lists: most will, for a small charge, and that removes the hardest cuts.

Tool	What it's for	Get it	About £
MEASURE AND MARK			
Tape measure (5 m) and a steel rule	every cut and every height	Buy	10–15
Combination square	squaring lines across boards and checking corners	Buy	10–15

Spirit level, 60 cm and 120 cm	levelling the base, the shelves and the table	Buy	15–30
Cross-line laser level	marking all the heights round the alcove in one go	Buy or borrow	30–60
Cable and pipe detector	checking every wall before you drill: non-negotiable	Buy	15–30
Pencil, masking tape, a marking knife	marking cuts (tape stops oak splintering)	Buy	5
CUT			
Track saw, or a circular saw with a straight-edge guide	the worktop, the table top and the long shelves	Hire or borrow	25–40 a day
Jigsaw	notches and the cable slot	Buy or borrow	40–70
Tenon saw (Japanese pull saw is easiest)	the tenons and the wedge kerfs	Buy	15–25
Sharp chisels, 6, 12 and 25 mm, and a mallet	cleaning out mortises and housings	Buy	30–50
DRILL AND FIX			
Cordless drill-driver (18 V)	pilot holes and screws; most of the job	Buy or borrow	60–120
SDS or hammer drill with masonry bits	plugs into the brick walls and the chimney breast	Hire or borrow	15–25 a day
Dowel jig and 10 mm brad-point bit	the dowels under the uprights and at the wrap ledge	Buy	20–35
Clamps: four 60 cm sash or F-clamps, four small	glue-ups and holding work while you fix it	Buy	40–60
Stud finder (if any wall is plasterboard)	finding studs to fix into	Buy	15
SHAPE AND FINISH			
Router (1/4 in is enough) with a straight bit and a guide	the housings, the LED grooves and the cable groove	Hire, borrow or buy	60–120
Random-orbit sander, 120 and 180 grit	smoothing the oak before oiling	Buy or borrow	40–70
Hardwax oil, lint-free cloths, a brush	finishing the oak	Buy	25–35
SAFETY			
Safety glasses, ear defenders, FFP2 dust masks	every cut, every sanding job; oak dust is an irritant	Buy	15–25
Work gloves; a first-aid kit	handling boards and sheet goods	Buy	10
A sturdy step ladder	the high shelf, the ceiling and the PC shelf	Borrow	—

Six techniques you'll use

1. **Fixing into brick.** Detect, mark, drill with the SDS on hammer setting, clear the dust, push in a brown (7 mm) wall plug and drive a 5 × 60 mm screw. In plasterboard, find a stud, or use heavy-duty plasterboard fixings rated for the load.

2. **Cutting oak cleanly.** Put masking tape over the cut line, score it with a knife, cut with the good face down (circular or track saw) or up (jigsaw). Clamp a straight edge as a guide.
3. **Routing a housing** (a shallow slot a shelf sits in). Clamp a straight batten as a guide, set the bit 4 mm deep, rout, then step down to 12 mm in passes. Practise on an offcut first.
4. **A wedged through-tenon.** Mark the 6 × 2.4 cm mortise on both faces of the upright and drill most of it out with a 20 mm bit from each side. Chisel it to the line. Saw the tenon on the shelf's end. Saw a kerf down its middle, glue it, tap it home, then drive a glued oak wedge into the kerf. When it's dry, saw it flush and plane it.
5. **Dowelling.** Use a dowel jig to drill matching 10 mm holes 3 cm deep in both parts, glue the dowels, clamp.
6. **Oiling.** Sand to 180 grit, wipe off the dust, rub on a thin coat of hardwax oil, wipe off the excess after 15 minutes, let it dry overnight, give it a second coat. Do the undersides too, so the boards don't cup.

The order of work

#	Stage	What to do	Check before moving on	Time
1	**Measure and count**	The alcove's width at the floor, 76, 134, 155 and 196 cm and at the ceiling: cut everything to the smallest. The floor's level across the alcove, and the chimney breast's flatness with a long level. The door's swing, and the sofa: overall width, depth with the return, back height. Count the LPs, singles and games (PS2, Xbox 360, Xbox One), and measure ten cases of each. Make a cardboard template of the alcove at worktop height: walls are rarely square.	Every figure within 1 cm of the plan. If not, send the numbers and the drawings are redone first.	2 h
2	**Network, stage 1**	Unplug everything from the Hub 5. Log in at 192.168.0.1 with the password on its base, then Modem Mode → Enable Modem Mode → Apply. Its page moves to 192.168.100.1. Connect a hub port to Deco A in bedroom 3 and set it up in the Deco app, reusing the old Wi-Fi name and password. If it shows no internet, try another hub port and restart the hub. Plug the new 8-port Switch 1 into Deco A's other port, then the PCs into Switch 1. Add Deco B in the living room and Deco C in the kitchen. For now they link over Wi-Fi. Speed-test each room, and check the Xbox's network settings show NAT: Open.	Internet in every room, and NAT: Open on the Xbox.	2 h
3	**Electrician**	Book one visit for all of it, and get a quote first. Two Cat6 runs from bedroom 3, across the landing and down through the airing cupboard: one to the alcove's right corner for Deco B, one on to the kitchen for Deco C. A spare pulled with each; the optional runs to the master and bedroom 2 if you want them. Swap the single socket on the side wall for a double at the same spot. Add a new double low on the side wall behind the sofa, for the PC and the rear speakers. Chase a 25 mm conduit into the side wall for the PC's leads, from behind the PC shelf to skirting height and behind the skirting to the base, with a draw wire left in it. Make good. LED-compatible dimmers for both ceiling points, both spot bars hung, and a fused spur for the LED drivers. Optional: a conduit to TV height for a future wall mount. Before they go, mark and photograph every cable route on the walls.	The certificate, photos of every route, and both ends of each Cat6 run labelled.	1–2 days (theirs)

4	**Paint**	Paint an A4 board in Card Room Green and look at it by day and with the LED strip on. Wash the skirting and door with sugar soap, sand to a key, and prime them. Two coats of emulsion on the walls, ceiling, alcove and chimney breast. Two coats of eggshell in the same green on the skirting, door, frame and radiator. Leave it a week to cure before fixing anything to it.	Two even coats, and no white showing at the edges or behind the radiator.	2–3 days, then a week to cure
5	**Mark the heights**	With the laser level: 73 and 76 for the worktop, 130 and 134 for the Series X shelf, 153 and 155.4 for the ledges, 192 and 196 for the high shelf. The uprights at 0–4.5 and 131–135.5 cm from the side wall. The dining wall's backboard at 51.3–73.3, centred on the breast. Run the cable detector along every line you will drill, and mark anything it finds.	Every line level end to end, and the uprights' lines plumb.	1 h
6	**Build the base**	Order the oak and ply from the cutting list (below the steps), a week or two ahead: most merchants cut to size for a small charge. Build the 9 cm plinth for bays 1–2, with the open bay for the hidden drawer under bay 2. Level it with shims. Assemble the two BESTÅ frames with their back panels left off and stand them on the plinth. Build bay 3 from 18 mm ply, 60 × 40 × 73, and bolt it to bay 2. Screw the 4 cm spacer blocks to the wall, push the base against them and fit the anti-tip brackets. Lay the flat two-compartment trunking in the channel behind (mains one side, data the other) and run the cables in.	The tops level within 1 mm across 186 cm, and nothing rocks.	1 day
7	**Fit the worktop**	Cut the worktop and the dining wall's backboard from the one 300 × 62 oak board, using the template from step 1. Scribe the ends to the walls; cut the TV slot over the channel and the grommet hole behind the deck. Oil it on every face, undersides and cut ends included. Screw it up through the frame tops in slotted holes, so the oak can move.	Tight to both walls, and the slot sits over the channel.	½ day
8	**Fit out the base**	Shelves on their pins at the fit-proof heights for the consoles. The game drawer in bay 1: runners on the frame's sides, then the box (cut list in the Media wall tab). The record trays in bay 3: runners at 2.3 and 37.0 cm, then each tray. The hidden plinth drawer under bay 2. Bay 2: the divider and the part shelf, then the sub on its riser, its front flush with the face frame.	Every drawer runs smoothly, and the two trays clear each other by 2 cm.	1–2 days
9	**Uprights and shelves**	On the bench: cut the housings and the mortises in both uprights, and the LED groove and cable groove. Dry-fit each upright with its shelves before any glue. Lay Deco B's leads in the groove behind the right upright. Stand the uprights on their dowels and fix the concealed brackets to the back wall. Fit the high shelf and the ledge, then the Series X shelf: wedge its tenons, and peg the housings.	Uprights plumb, shelves level, and the TV slides in with 2 cm each side.	2–3 days
10	**Side wall and PC shelf**	Screw the oak cleat for the wrap ledge and the PC shelf to the side wall. Dowel the wrap ledge to the left upright's front edge. Fit the concealed PC shelf brackets, then the shelf. Screw the keyboard tray's housing under it and fit its runners. Fit the brush grommet and pull the PC's leads down the chased conduit with the draw wire.	One level line with the ledge over the TV, and the PC shelf holds your weight leaning on it.	1 day

11	**The dining wall**	Screw the backboard to the breast, plugs every 20 cm. Fit the two Hebgo brackets on its face, 80 cm apart, arm tops level with its top. Screw the lift-off hinges to the backboard's top edge and the table's back edge, all the same hand. On the bench: oil the table's underside and set the four brass inserts for the legs. Hang the table and fit the two turn buttons above it.	It folds up and down without binding, both brackets click and lock, and it doesn't move when you lean on it.	1 day
12	**Lighting**	LED strip into each upright's groove, wired to its driver on the fused spur. The ceiling strip through its notches in the uprights; the bias light on the back of the TV. Aim both spot bars (the table in the Paint and light tab says which head goes where). The lamp by the sofa, with a 2700K bulb.	From the sofa, no spot head shows in the screen or shines in your eyes.	½ day
13	**Connect and finish**	TV inputs: HDMI 1 the PC, HDMI 2 the soundbar (eARC), HDMI 3 the Series X, HDMI 4 the HDMI switch (Xbox One, 360, PS3, RetroTINK). The PS2 by component into the RetroTINK and by optical to the soundbar; the PS1 by RGB SCART. The deck to the preamp, the preamp to the soundbar's AUX; set the tracking weight; soundbar in Music mode for vinyl. Network stage 2: plug each Cat6 run into its Deco, check Deco B and C show "Ethernet", and give the Series X and PC high priority. Label both outlets of the double socket and each HDMI switch input; set the sub and rear levels from the sofa.	Everything on the network's "done when" list, and every source plays through the soundbar.	1 day

Roughly 12–15 working days spread over a few weekends, plus the electrician and the paint drying. Steps 6–11 can be split: the base and dining wall first, the oak joinery once you're confident.

Where beginners usually go wrong

- **Cutting the worktop to the tape instead of to a template.** Make a cardboard template of the alcove first; walls are rarely square.
- **Skipping the dry-fit.** Gluing a tenon that doesn't fit is the one mistake you can't undo.
- **Oil on one side only.** The board cups within a week.
- **Loading the trays before the anti-tip brackets go in.** A full tray weighs about 32 kg.
- **Folding the table up with something on it.** Clear it first, every time, and keep both turn buttons turned when it's up.

The cutting list

Every made part in one list, grouped by what it's cut from, for the timber merchant. Sizes are finished sizes in cm, length × width × thickness. The cut lists in the other sections say how each part joins; this list is for ordering.

Solid oak, hardwax oil

Ask for it planed all round; scribe the uprights and worktop on site

Part	Size (cm)	Qty	Notes	Section

Uprights	159 × 30 × 4.5	2	Buy 165 long and scribe to the ceiling; housings, mortises, LED and cable grooves cut on the bench	Media wall
High shelf, long length	128.9 × 30 × 4	1	Housed 1.2 into both uprights and pegged	Media wall
High shelf, short length	51.7 × 30 × 4	1	Housed into the right upright; cleat on the reveal	Media wall
Series X shelf	55 × 30 × 4	1	Two through-tenons 6 × 2.4, wedged flush	Media wall
Record ledge · wrap ledge	128.9 × 7 × 2.4 · 134 × 7 × 2.4	1 · 1	Housed and pegged · on a cleat, dowelled to the left upright	Media wall
Ledge lips	128.9 × 1.5 × 0.9 · 134 × 1.5 × 0.9	1 · 1	Glued along the ledges' front edges	Media wall
PC shelf	82 × 28 × 4	1	Oak cleat and two concealed brackets	Media wall
Keyboard tray housing	70 × 28 × 7, sides 7 × 2	1	Screwed under the PC shelf	Media wall
Worktop	186 × 46 × 3	1	From one 300 × 62 full-stave worktop; chamfered front	Media wall

Dining backboard	110 × 22 × 3	1	From the same worktop's offcut	Dining table
Face frame	186 × 73 overall, 2 thick, stiles about 5 wide	1 set	End stiles scribed to the walls	Media wall
Table top, centre	101 × 80 × 2.7	1	Edge-glued staves: 96 between the shoulders, plus a 2.5 tongue each end	Dining table
Breadboard ends	80 × 7 × 2.7	2	Pegged with three 10 mm pegs each, in slotted holes	Dining table
Leaf	60 × 80 × 2.7	1	Two alignment dowels and two table clips	Dining table
Legs (oak option)	71 × 4.5, tapering to 3	8	M8 hanger bolts; or buy the Emuca steel legs instead	Dining table
Turn buttons	6 × 2.5 × 2.2	2	On a single screw each	Dining table
Drawer fronts	game drawer 56.8 × 6.5 × 2 · plinth drawer 54 × 6 × 2	1 · 1	Finger pull cut underneath	Media wall
Record tray lips	about 56 × 5 × 1.5	2	Cut to the face frame's opening, less 3 mm	Media wall
Record tray dividers	38.2 × 8 × 0.6	2	Between the LP and singles troughs	Media wall
Follower blocks · Xbox One lift block	19 × 12 × 2 · 15 × 12 × 2	2 · 1	Offcuts, felted underneath	Media wall
Wedges · pegs · dowels	6 × 3, tapering · 10 mm dowel, 5 long · 10 mm dowel	2 + spares · 6 + 6 · 8	Glued	Media wall

Birch ply, 18 mm

Part	Size (cm)	Qty	Notes	Section
Bay 3 sides	73 × 40	2	The record carcass, standing on the floor	Media wall
Bay 3 top and bottom	56.4 × 40	2	Between the sides; back left open to the channel	Media wall
Plinth rails	120 × 9	2	Front and back of the ladder frame under bays 1–2	Media wall
Plinth cross-pieces	34.4 × 9	3	At both ends and under the bay 1–bay 2 joint, so the bay under bay 2 stays clear for the drawer	Media wall

Birch ply, 12 mm

Part	Size (cm)	Qty	Notes	Section
Game drawer sides	35 × 15.5	2		Media wall
Game drawer back · front panel	51.9 × 15.5 · 51.9 × 6	1 · 1		Media wall
Record tray bases	53.9 × 40	2	The sides and ends stand on them	Media wall
Plinth drawer sides · ends	34 × 4.4 · 49.6 × 4.4	2 · 2	A 52 × 34 × 5 box	Media wall
Bay 2 divider · part shelf	35.3 × 38 · 32.2 × 36	1 · 1	The sub's column on the left; the PS2 over the PS1 on the right	Media wall
Keyboard tray	54 × 26	1	Oak lip on its front edge	Media wall

Birch ply, 9 mm

Part	Size (cm)	Qty	Notes	Section
Game drawer base	52.5 × 34.4	1	In a 6 mm groove	Media wall
Record tray sides · ends	40 × 8 · 52.1 × 8	4 · 4	Two of each per tray; new in v13.3: sides set at 8 cm tall, room for the runners	Media wall

Birch ply, 6 mm

Part	Size (cm)	Qty	Notes	Section
Game drawer cross divider	51.9 × 9	1	13 cm back from the front	Media wall
Game drawer bin walls · bin floors	12.6 × 9 · 16 × 12.6	2 · 2	Floors raked so the cases lean about 13°	Media wall
Plinth drawer base	52 × 34	1	Screwed under the box	Media wall

Hardware

Runners are sold in pairs

Part	Size (cm)	Qty	Notes	Section
Record tray runners	400 mm full extension, 68–70 kg a pair	2 pairs	Screwed straight into bay 3's sides	Media wall
Game drawer runners	350 mm full extension, soft-close	1 pair	Accuride DZ3832EC or Screwfix 964VN	Media wall
Plinth drawer and keyboard tray runners	300 mm full extension · 300 mm light duty	1 pair · 1 pair		Media wall
Häfele Hebgo folding brackets	287.43.428, 580 arm	2	500 kg a pair	Dining table
Lift-off hinges	Union PowerLoad 100 × 88, all the same hand	3		Dining table
Brass M8 threaded inserts	for the legs	8	Four in the top's breadboard ends, four in the leaf	Dining table
Sub isolation riser	dense foam or sorbothane, about 22 × 28 × 3	1	Under the sub in bay 2	Media wall
Concealed brackets	for the uprights; two for the PC shelf, 40 kg+ a pair	1 set		Media wall

Step by step, in full

Step 1 · Measure and count · You · 2 h

Tools

- Tape measure
- Spirit level, 120 cm
- Notebook or the Questions tab

1. The alcove's width at the floor, 76, 134, 155 and 196 cm and at the ceiling: cut everything to the smallest.
2. The floor's level across the alcove, and the chimney breast's flatness with a long level.
3. The door's swing, and the sofa: overall width, depth with the return, back height.
4. Count the LPs, singles and games (PS2, Xbox 360, Xbox One), and measure ten cases of each.
5. Make a cardboard template of the alcove at worktop height: walls are rarely square.

Check before moving on: Every figure within 1 cm of the plan. If not, send the numbers and the drawings are redone first.

Step 2 · Network, stage 1 · You · 2 h

Buy first: 1 Network, now (the buy list).

Parts

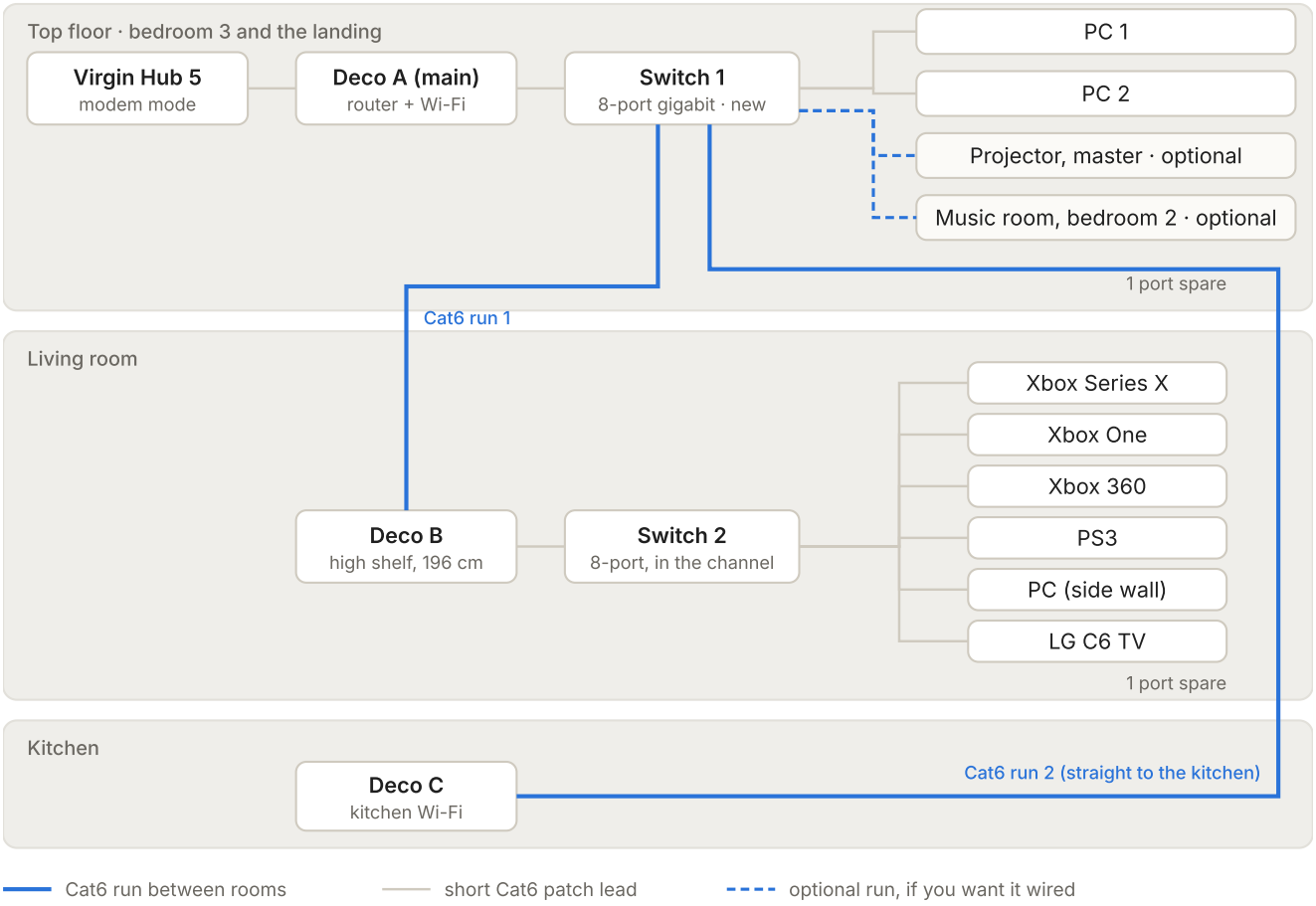
- Two TP-Link TL-SG108 8-port switches
- Cat6 patch leads, 0.5–2 m (11)

Tools

- A phone with the Deco app
- A laptop for the hub's settings page

Every floor has its own Cat6 link back to bedroom 3

Deco A and Switch 1 live with the Hub 5 and the gaming PCs in bedroom 3; two spare ports can wire the projector and the music room



1. Unplug everything from the Hub 5. Log in at 192.168.0.1 with the password on its base, then Modem Mode → Enable Modem Mode → Apply. Its page moves to 192.168.100.1.
2. Connect a hub port to Deco A in bedroom 3 and set it up in the Deco app, reusing the old Wi-Fi name and password. If it shows no internet, try another hub port and restart the hub.
3. Plug the new 8-port Switch 1 into Deco A's other port, then the PCs into Switch 1.
4. Add Deco B in the living room and Deco C in the kitchen. For now they link over Wi-Fi.
5. Speed-test each room, and check the Xbox's network settings show NAT: Open.

Check before moving on: Internet in every room, and NAT: Open on the Xbox.

Step 3 · Electrician · Electrician · 1–2 days (theirs)

Buy first: 2 Electrician and cabling (the buy list).

Parts

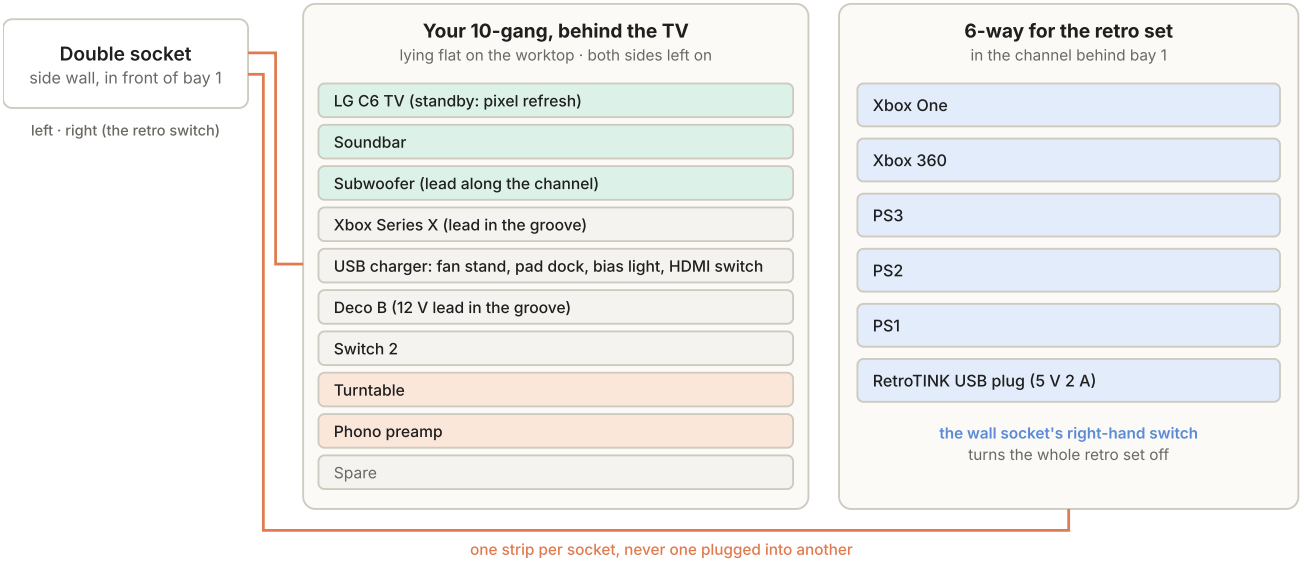
- Cat6 solid pure copper (not CCA), 2–4 runs
- 25 mm oval conduit and a draw wire, about 4 m
- Two LED-compatible dimmers
- Both spot bars and their bulbs (from the lighting phase, so the electrician can hang them)

Tools

- Nothing: it's the electrician's visit. Have the wall marks and this plan ready

Power map: your 10-gang plus a 6-way for the retro set

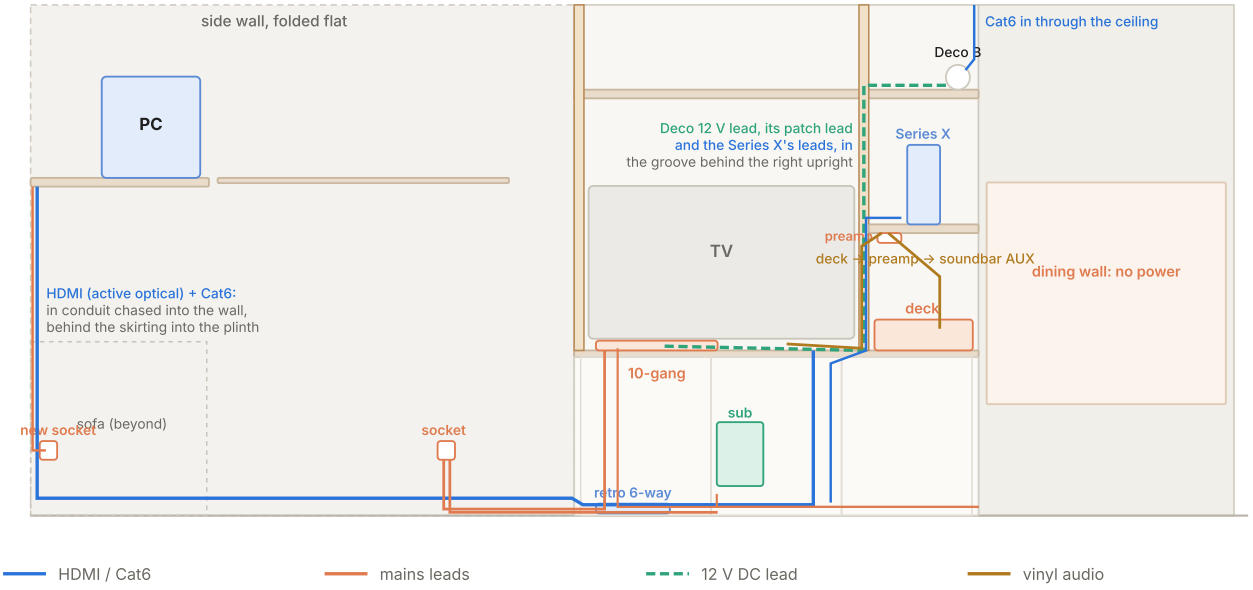
Everything modern on the left outlet, always on; the five retro consoles and the RetroTINK on the right, switched at the wall



- The socket on the side wall is a single today: the electrician swaps it for a double, at the same spot.
- The retro switch moves to the wall: with the TV 2 cm from the left upright, the 10-gang's own switches can't be reached any more.
- PC and rear speakers: a new double socket low on the side wall behind the sofa; the PC (500–850 W) stays off these strips.
- The sub plugs into the 10-gang: its lead drops through the worktop's slot to the back of bay 2. Both spot bars use existing ceiling points.
- Load check: the 10-gang about 0.7 kW with everything on, the 6-way up to about 0.8 kW with a launch PS3 running; each well under 13 A (3 kW).
- Check the 10-gang is surge-protected; if not, use a surge-protected one for the TV and consoles. Label both outlets.

Cable routes: everything ends in the channel behind the base

Front view, schematic, with the side wall folded flat to the left so the PC's route shows; the sub sits under the dining wall



1. Book one visit for all of it, and get a quote first.
2. Two Cat6 runs from bedroom 3, across the landing and down through the airing cupboard: one to the alcove's right corner for Deco B, one on to the kitchen for Deco C. A spare pulled with each; the optional runs to the master and bedroom 2 if you want them.
3. Swap the single socket on the side wall for a double at the same spot. Add a new double low on the side wall behind the sofa, for the PC and the rear speakers.
4. Chase a 25 mm conduit into the side wall for the PC's leads, from behind the PC shelf to skirting height and behind the skirting to the base, with a draw wire left in it. Make good.
5. LED-compatible dimmers for both ceiling points, both spot bars hung, and a fused spur for the LED drivers. Optional: a conduit to TV height for a future wall mount.
6. Before they go, mark and photograph every cable route on the walls.

Check before moving on: The certificate, photos of every route, and both ends of each Cat6 run labelled.

Step 4 · Paint · You · 2–3 days, then a week to cure

Buy first: 3 Paint (the buy list).

Parts

- Card Room Green emulsion and eggshell, primer and undercoat
- Sugar soap, sanding pads, woodwork primer

Tools

- Roller, tray and extension pole
- 25 and 50 mm brushes
- Dust sheets, masking tape
- Step ladder

1. Paint an A4 board in Card Room Green and look at it by day and with the LED strip on.
2. Wash the skirting and door with sugar soap, sand to a key, and prime them.
3. Two coats of emulsion on the walls, ceiling, alcove and chimney breast.
4. Two coats of eggshell in the same green on the skirting, door, frame and radiator.

5. Leave it a week to cure before fixing anything to it.

Check before moving on: Two even coats, and no white showing at the edges or behind the radiator.

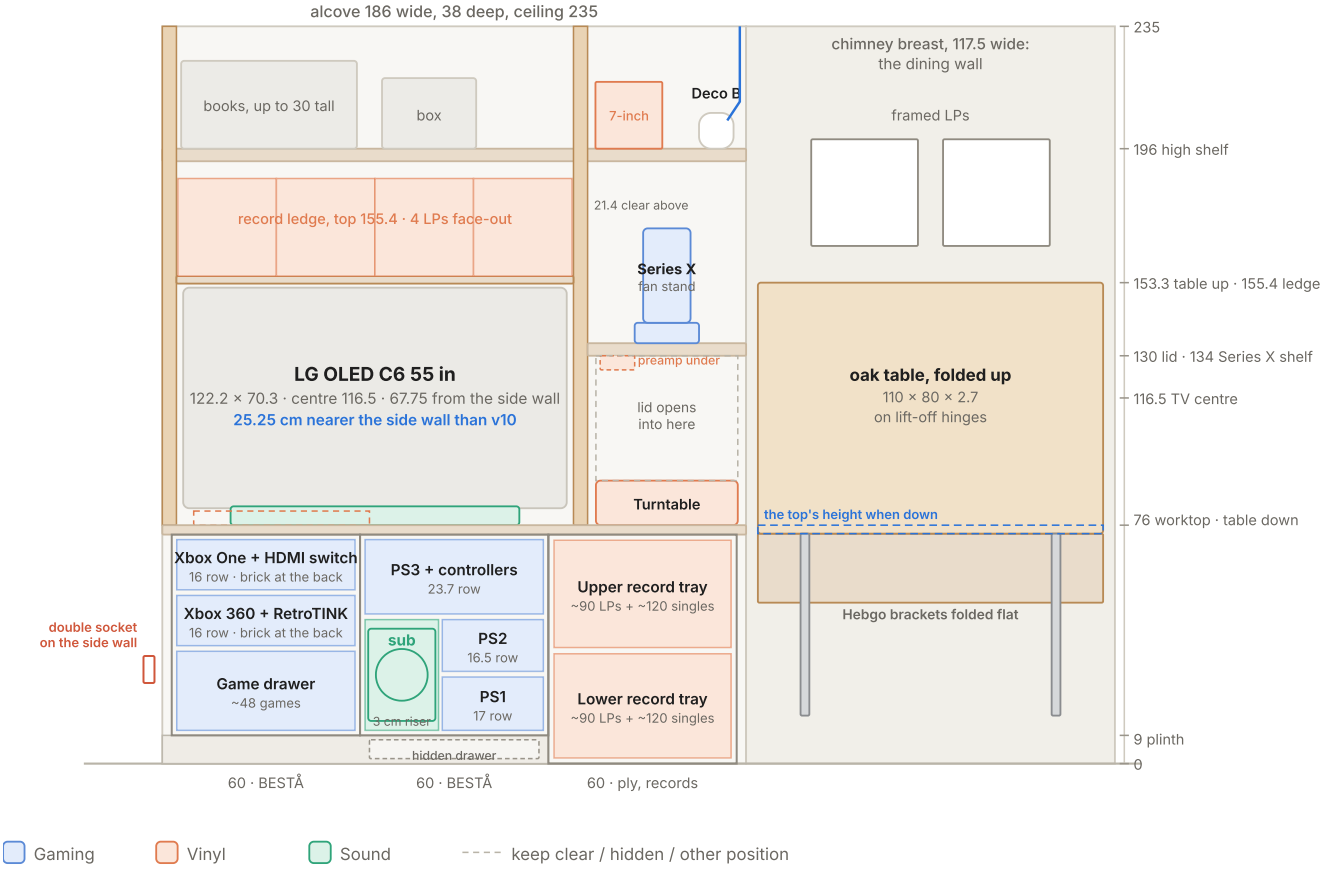
Step 5 · Mark the heights · You · 1 h

Tools

- Cross-line laser level
- Tape measure
- Pencil and masking tape
- Cable and pipe detector
- Spirit level, 120 cm

The media wall and the dining wall from the front, to scale

The TV between oak uprights nearer the side wall, the turntable zone on its right, three bays below; the fold-down table on the chimney breast. Heights in cm



Set up a datum

1. Find the floor's highest point across the alcove (you measured its level in step 1). Every height in this plan is measured from there.
2. Stand the laser in the middle of the room, level it, and mark a line at exactly 100 cm above that high point on the alcove's back wall, both side walls and the chimney breast.
3. Measure every other height up or down from the 100 line, not up from the floor: the floor is the thing that isn't level.

The alcove's heights

4. Put a strip of masking tape down each side wall and the back wall of the alcove and mark these heights on it:
5. 73: the tops of the frames and bay 3 (the worktop's underside).
6. 76: the worktop's top.
7. 130 and 134: the Series X shelf, underside and top.
8. 153 and 155.4: the record ledge and the wrap ledge, underside and top.
9. 192 and 196: the high shelf, underside and top.
10. Carry the 153 and 155.4 lines along the side wall to 250 cm out from the alcove's back wall, for the wrap ledge and the PC shelf.

Across the back wall

11. Mark the uprights' positions from the side wall: the left one 0–4.5, the right one 131–135.5. Turn the laser to its vertical line and draw both lines from the worktop line to the ceiling.
12. Mark the base from the side wall along the floor: filler 0–3, bay 1 3–63, bay 2 63–123, bay 3 123–183, filler to 186.

On the chimney breast

13. Find the breast's centre: 58.75 cm from its alcove-side corner. Draw it plumb from the floor to 170.
14. Mark the backboard: 55 cm either side of the centre line, from 51.3 to 73.3 high.
15. Mark the two bracket centres 40 cm either side of the centre line.
16. Mark the folded table's top edge at 153.3, and the framed LPs' bottom edges at 165.

Check for cables

17. Run the detector along every line where you'll drill: the cleats, the spacer blocks, the upright brackets, the backboard and the PC shelf.
18. Mark anything it finds in red, and compare with the electrician's photos from step 3. Move a fixing 5 cm rather than drill near a cable.

Check before moving on: Every line level end to end, and the uprights' lines plumb.

Step 6 · Build the base · You · 1 day

Buy first: 4 Base unit (the buy list).

Parts

- Two IKEA BESTÅ frames 60 × 40 × 64
- 18 mm birch ply: bay 3 sides 73 × 40 (2), top and bottom 56.4 × 40 (2); plinth rails 120 × 9 (2), cross-pieces 34.4 × 9 (3)
- Four 4 cm spacer blocks (ply offcuts, 4 × 10 × 1.8)
- Anti-tip brackets: two per bay (6)
- Flat two-compartment trunking, about 2 m
- Plastic shims
- Four connector bolts (furniture joint connectors)

Hardware and fixings

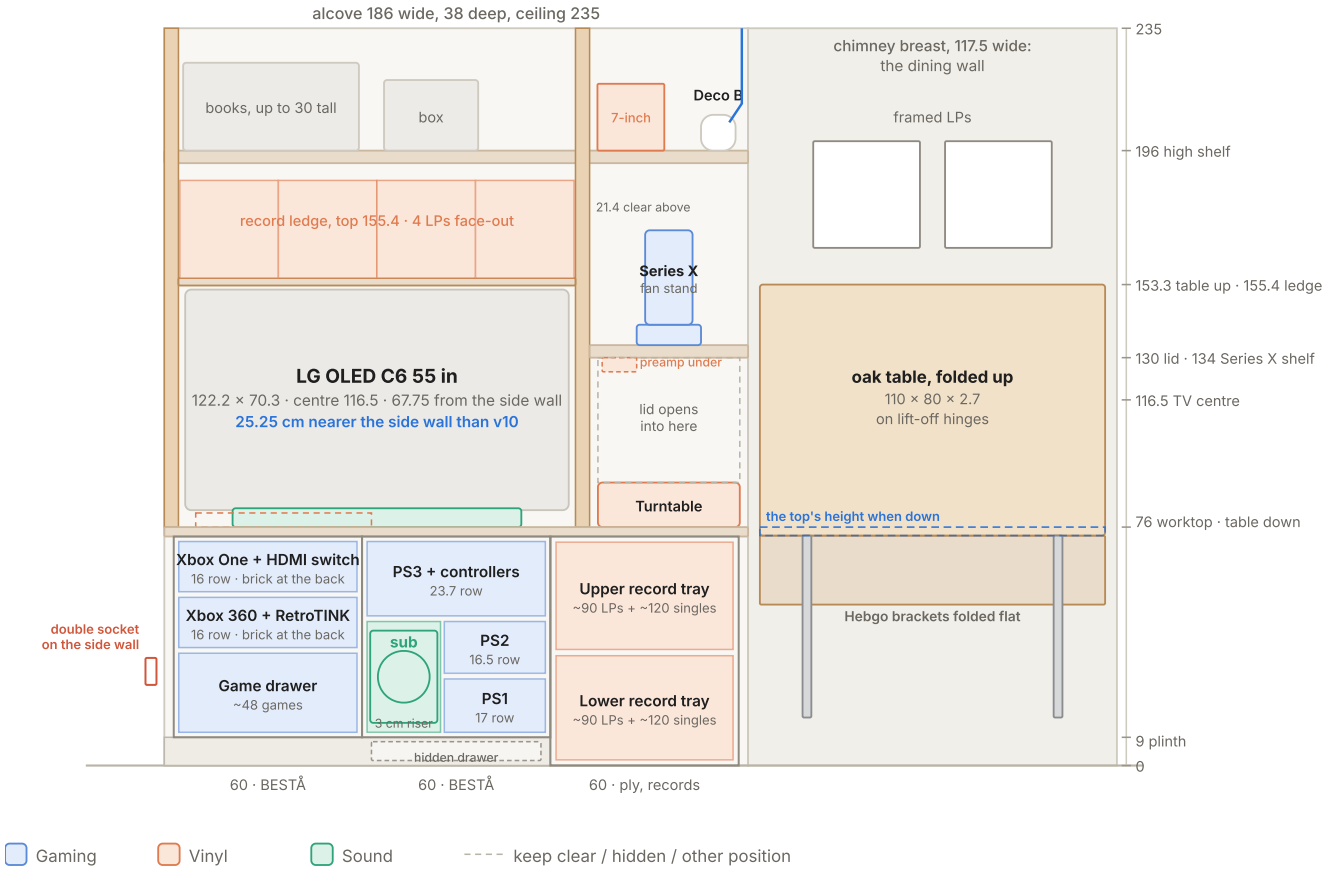
- 4 × 40 and 4 × 50 mm wood screws
- 5 × 60 mm screws and brown wall plugs
- Wood glue

Tools

- Drill-driver with pilot and countersink bits
- SDS drill with a 7 mm masonry bit
- Spirit level, 60 and 120 cm
- Combination square
- Clamps
- Tape measure

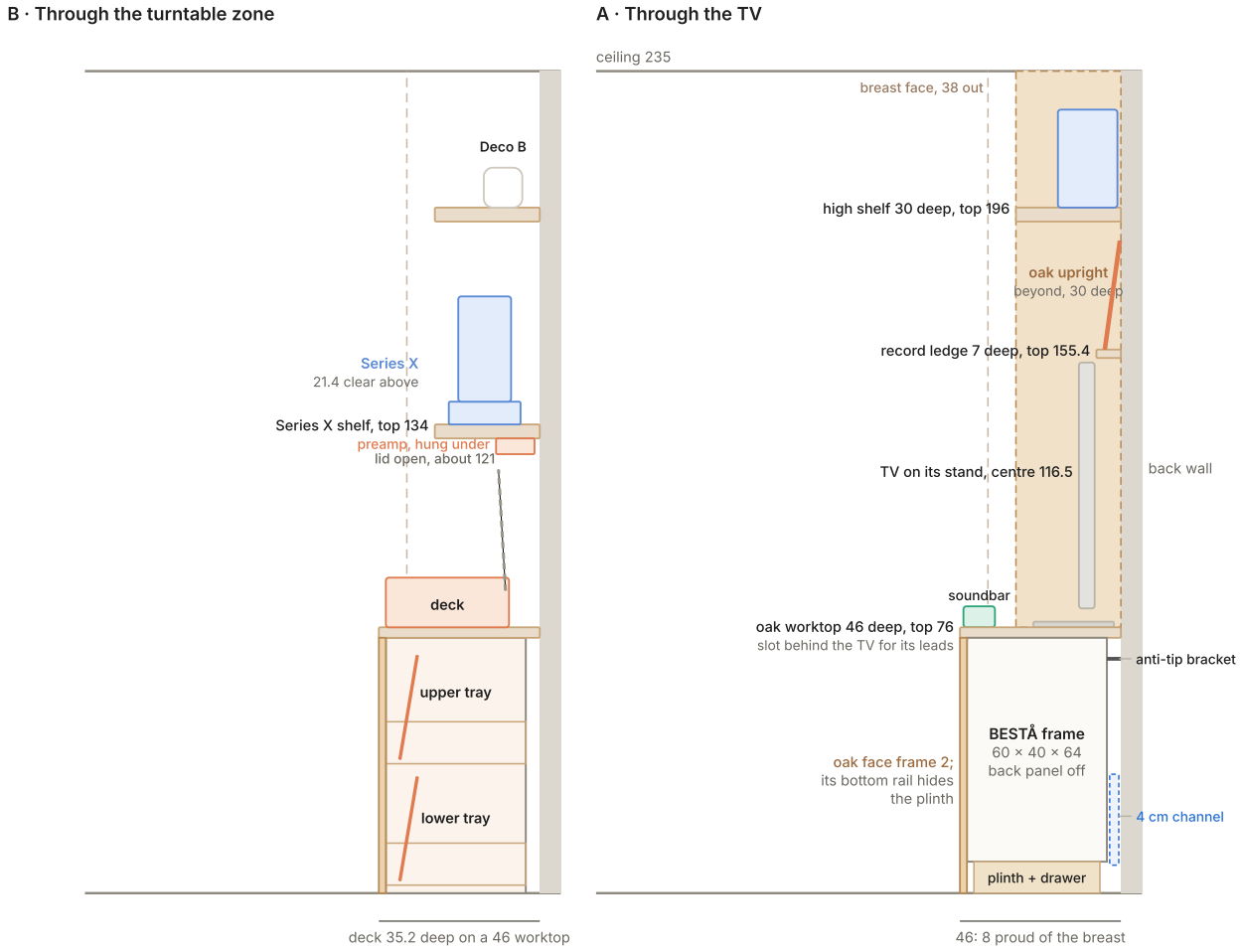
The media wall and the dining wall from the front, to scale

The TV between oak uprights nearer the side wall, the turntable zone on its right, three bays below; the fold-down table on the chimney breast. Heights in cm



How the wall goes together, layer by layer

Side sections to scale, room on the left. Dashed: the upright and the breast's face beyond the cut, and the deck's lid open



The plinth under bays 1 and 2

1. Lay the two 120 × 9 rails on edge, 38 cm apart outside, and stand the three 34.4 × 9 cross-pieces between them: one at each end and one in the middle, at 60.
2. Pre-drill, glue and screw through the rails into each cross-piece end with two 4 × 40 screws.
3. Measure both diagonals: when they're equal the frame is square. The plinth is 120 × 38 × 9. The opening between 60 and 120 stays clear: it's the hidden drawer's bay.
4. Stand it in the alcove 3 cm from the side wall and 4 cm off the back wall. Level it with shims, side to side and front to back.

The BESTÅ frames

5. Assemble both frames to IKEA's instructions, but leave the back panels off: the open backs are how the cables and the sub's port breathe.
6. Join the two frames side by side with IKEA's connectors.
7. Stand them on the plinth, their backs flush with its back edge, so their fronts sit 2 cm proud of the plinth (the face frame covers the step).

Bay 3, the record carcass

8. While the sides are flat, mark the runner lines on their inner faces: 2.3 and 37.0 cm up from the bottom edge, for the two trays' undersides.
9. Stand the bottom panel between the two sides, flush with their bottom edges, and the top panel flush with their top edges. Pre-drill, countersink, glue and screw with four 4 × 50 screws a joint.

10. Check the diagonals for square. The carcass is 60 × 40 × 73, open at the back.
11. Stand it in its place, 123–183 from the side wall, its top level with the frames' tops at 73. Shim under it if the floor drops away.
12. Clamp it to bay 2 and fit two connector bolts at the front and two at the back.

Fix it to the wall

13. Screw a 4 cm spacer block to the back wall behind the top of each bay, at 68 high, with plugs.
14. Push the whole base back against the blocks.
15. Fit two anti-tip brackets per bay through the blocks into the wall. Do this before anything heavy goes in: a full record tray pulled out is about 32 kg in front of the unit.
16. Lay a level across all three tops: within 1 mm over 186 cm. Shim and re-check.

The cable channel

17. Lay the flat two-compartment trunking on the floor in the 4 cm channel behind the base: mains on the wall side, data on the other.
18. Run the leads in now: the two mains leads from the side-wall socket enter through the plinth's left end, the Cat6 and Deco B's lead come down the right-hand corner.
19. Keep the channel behind bay 2 empty: the sub's port breathes into it. Switch 2 and the retro 6-way stand on edge behind bay 1.

Check before moving on: The tops level within 1 mm across 186 cm, and nothing rocks.

Step 7 · Fit the worktop · You + a helper · ½ day

Buy first: 5 Oak joinery (the buy list).

Parts

- One solid oak worktop, 300 × 62 × 3, full stave
- A 60 mm brush strip and a 35 mm brush grommet

Hardware and fixings

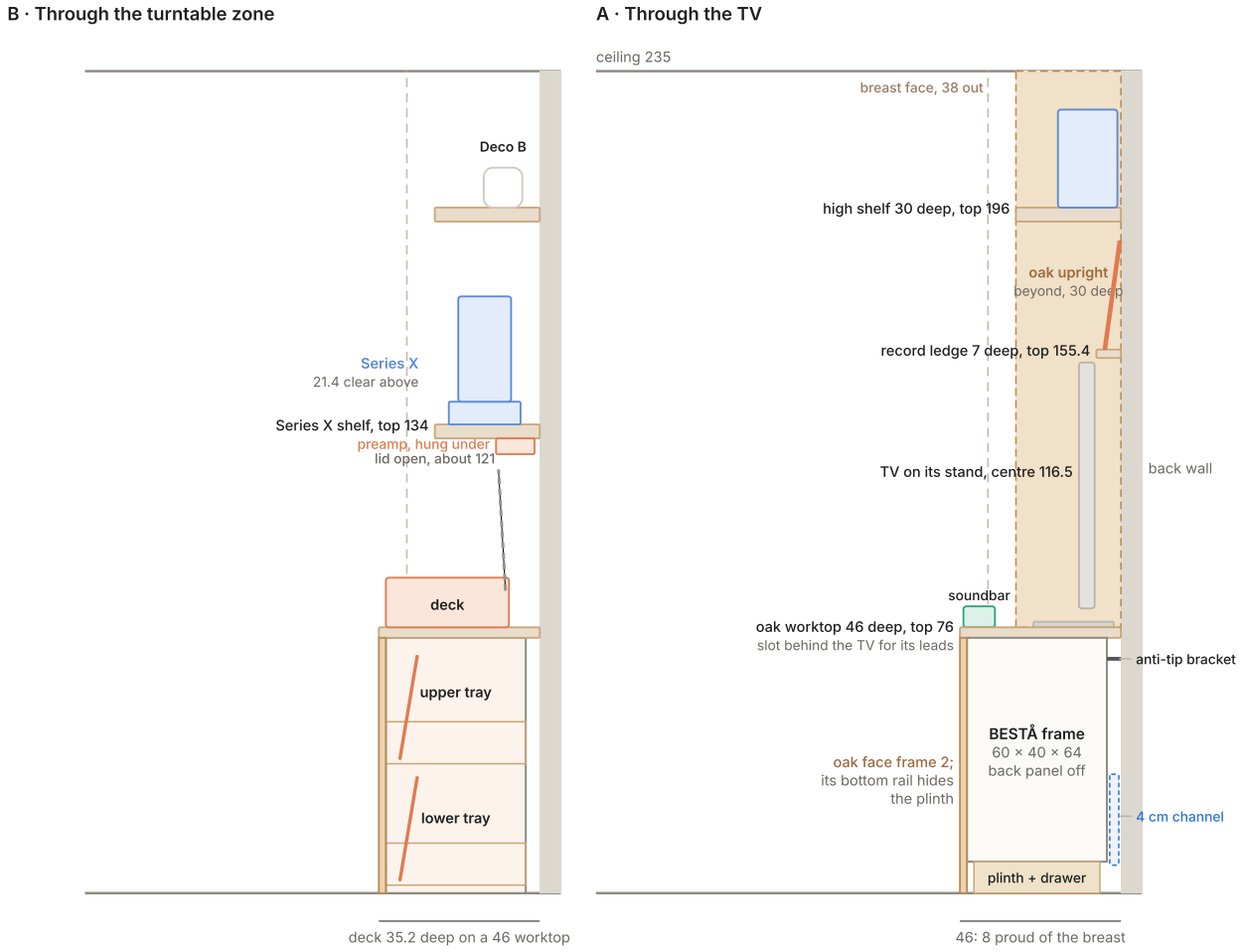
- 4 × 30 mm screws with washers (9)
- Hardwax oil

Tools

- Hardboard or cardboard, 190 × 50, for the template
- Compass or a pencil on a block, for scribing
- Track saw or circular saw with a guide
- Jigsaw
- Router with a straight bit
- Random-orbit sander, 120 and 180 grit
- Drill-driver
- A helper

How the wall goes together, layer by layer

Side sections to scale, room on the left. Dashed: the upright and the breast's face beyond the cut, and the deck's lid open



Make a template

1. Cut the hardboard to about 190 × 50 and push it into the alcove at worktop height, its back edge against the back wall.
2. Set the compass to the widest gap at one end and run it down the side wall, so the pencil copies the wall's line onto the board. Cut to the line. Do the other end the same way.
3. Mark the front edge at 46 cm from the back wall all along. The template is now the worktop's exact shape.

Cut the oak

4. Lay the board best face up, grain running along its length. Lay the template on it and draw round it.
5. Cut just outside the line with the track saw, then sand or plane to the line.
6. From the offcut, cut the dining wall's backboard: 110 × 22 × 3.
7. Chamfer the worktop's top front edge with a 3 mm chamfer, sanded by hand.

Cut the openings

8. The TV slot: 3 cm wide, from 1 to 4 cm in from the back edge, running 27.75–107.75 from the side wall. Clamp a batten as a guide and rout it in passes, or drill each end and jigsaw between. Push the brush strip in.
9. The grommet hole behind the deck: 35 mm, its centre 2.5 cm from the back edge and 160.75 from the side wall.

Finish and fix

10. Sand to 120, then 180. Oil every face: top, underside, both ends and the cut edges of the slot. Two coats, a night between them.
11. Two of you lift it in and push it back to the wall. It should touch both side walls along their length.
12. Inside each bay, drill three 5 mm holes up through the top panel, elongated front to back to about 12 mm long, so the oak can move with the seasons.
13. Screw up through them into the worktop with 4 × 30 screws and washers. They must not come out through the top: check the length against a scrap.

Check before moving on: Tight to both walls, and the slot sits over the channel.

Step 8 · Fit out the base · You · 1–2 days

Parts

- Four BESTÅ shelves 56 × 36
- Game drawer: 12 mm ply sides 35 × 15.5 (2), back 51.9 × 15.5, front panel 51.9 × 6; 9 mm base 52.5 × 34.4; 6 mm divider 51.9 × 9, bin walls 12.6 × 9 (2), bin floors 16 × 12.6 (2); oak lip 56.8 × 6.5 × 2; follower blocks
- Record trays (×2): 12 mm base 53.9 × 40; 9 mm sides 40 × 8 and ends 52.1 × 8; 6 mm oak divider 38.2 × 8; oak lip about 56 × 5 × 1.5; felt
- Plinth drawer: 12 mm sides 34 × 4.4 and ends 49.6 × 4.4, 6 mm base 52 × 34, oak front 54 × 6 × 2
- Bay 2: 12 mm divider 35.3 × 38, part shelf 32.2 × 36, the sub's isolation riser about 22 × 28 × 3
- The oak face frame, 2 cm thick: stiles about 5 wide, rails; 186 × 73 overall

Hardware and fixings

- Runners: two pairs of 400 mm (68–70 kg), one pair of 350 mm soft-close, one pair of 300 mm
- 3.5 × 16 mm screws for the runners (every hole)
- Four shelf pins for the part shelf
- Two small steel angle brackets for the divider
- Panel pins, wood glue, felt

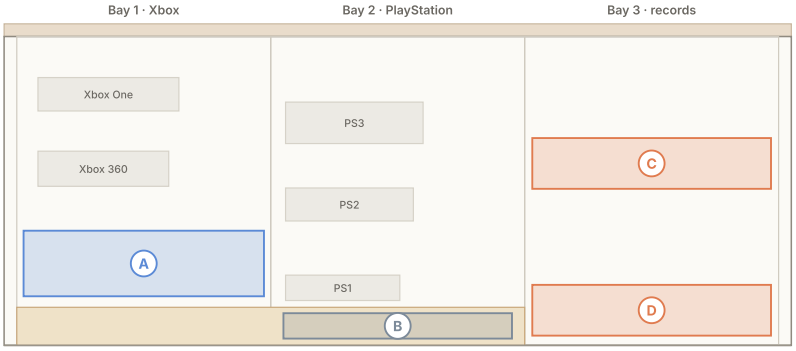
Tools

- Drill-driver
- Router with a 6 mm straight bit (the game drawer's base groove)
- Combination square
- Spirit level
- Clamps
- Mitre saw or tenon saw

The drawer layouts: where each drawer is and what goes in it

Four drawers in the alcove base, all to scale: A the game drawer, B the hidden plinth drawer, C and D the two face-out record trays

Where each drawer is: the alcove base from the front



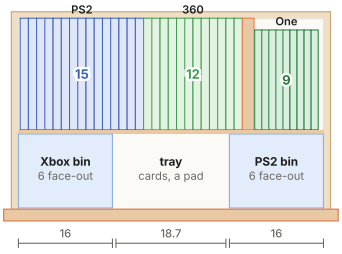
- **A · Game drawer: bay 1, bottom row (11.6–27.1 up)**
- **B · Plinth drawer: hidden under bay 2 (1.6–7.6 up)**
- **C · Upper record tray: bay 3 (37–69.1 up)**
- **D · Lower record tray: bay 3 (2.3–34.4 up)**

Open one at a time: a full record tray is about 32 kg.
Anti-tip brackets fix the base to the wall first.
Every front: a plain oak lip, finger pull underneath.

186 wide: 3 + 60 + 60 + 60 + 3 · worktop at 76 · front view, 3.6 px per cm

A · Game drawer

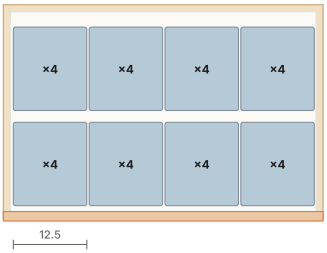
box 54.3 × 35 · 350 mm soft-close runners · rolls out 30



back: 36 spine-up, titles on top (19 cm cases on edge)
front: 12 face-out · about 48 games in all

B · Hidden plinth drawer

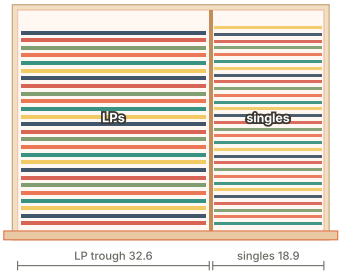
box 52 × 34 × 5 inside · 300 mm runners · finger pull



32 PS1 games flat: 8 stacks of 4 jewel cases
or remotes, spare pads, cables and manuals

C and D · Record trays (the two are the same)

box 53.9 × 40 · 400 mm runners, 68 kg a pair



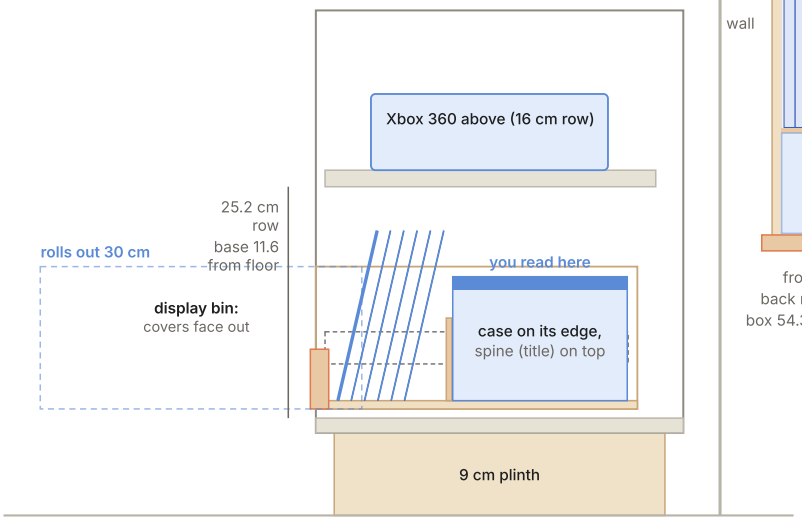
every sleeve face-out, leaning back 10°: flip like a shop
~90 LPs + ~120 singles each · two trays

Plans drawn from above at 5 px per cm, front (the room) at the bottom; the elevation at 3.6 px per cm. Sizes: BESTÅ frame 60 × 40 × 64 (IKEA);
game cases 19 × 13.5 × 1.4 and 17 × 13.5 × 1.2; jewel case 14.2 × 12.5 × 1; sleeves 31.4 and 18.3; runners Emuca 400 mm 68 kg, Accuride 350 mm (see the fit proof).

The game drawer: covers on show at the front, every title readable from above at the back

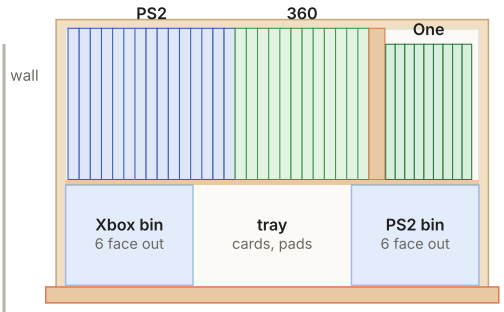
Bay 1, bottom row. Section left, plan right. Nothing stands behind anything else

Side section through bay 1



box 35 cm deep on 350 mm full-extension soft-close runners
5 cm free behind the box, then the 4 cm cable channel
bins: 19 cm cases leaning 13°, about 18.5 cm tall

Top view, drawer pulled out

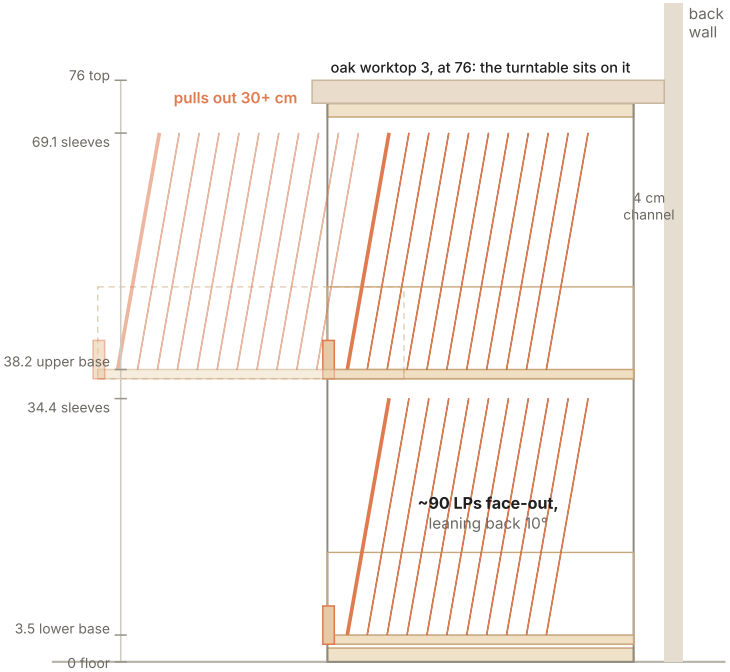


front: oak lip with finger pull, matching the record drawer
back row: 15 PS2 + 12 Xbox 360 + follower block + 9 Xbox One
box 54.3 × 35 cm outside · 51.9 cm inside · 16 cm bins, 18.7 cm tray

The record trays: every LP and single face-out, two tiers in bay 3

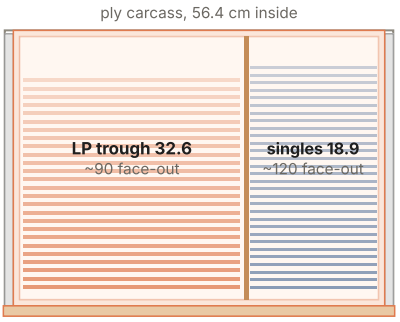
Bay 3 of the alcove base, right under the turntable. Section left, plan right. Pull a tray and flip through the covers like a record shop browser

Side section through bay 3



carcass 18 mm birch ply, 60 × 40 × 73, back open to the channel · trays 40 deep on 400 mm runners
12 mm base, 9 mm sides, 5 cm oak lip · 2.6 and 2.1 cm finger room (top panel at 71.2)

Top view of one tray

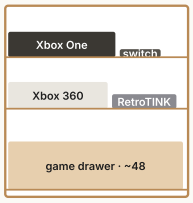


tray 53.9 × 40 outside, 52.1 × 38.2 inside
9 mm sides · 6 mm oak divider · 5 cm oak lip
about 32 kg full: open one tray at a time

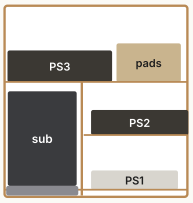
Storage options: what fits each space, to scale

Front views and views from above of every storage space in the wall, with counts, and where the overflow would go

Bays 1 and 2: inside each BESTÅ frame
56.8 wide × 40 deep inside; rows as planned, front view



Bay 1 · Xbox



Bay 2 · PlayStation + sub

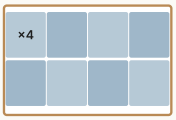
Swap any 16 cm row for two KUGGIS boxes, 26 × 35 × 15, side by side (52 of 56.8 wide).

Bay 3: each record tray, from above
53.9 × 40 outside, two trays; everything face-out


LP trough 32.6 · singles 18.9

≈ 90 LPs + 120 singles a tray
× 2 trays = 180 LPs, 240 singles
sleeves lean back 10°, flip like a shop
thick outer sleeves: widen the LP trough to 33.4; singles drop to 18.1

Hidden plinth drawer, from above
box 52 × 34 × 5 inside, under bay 2; finger pull, not push-to-open



32 PS1 games flat
jewel cases 14.2 × 12.5 × 1, 8 stacks of 4
or: remotes, spare pads, cables, manuals

Series X shelf, front view
50.5 wide, 134 to the high shelf at 192: 58 clear



15 air each side, 21.4 above
Microsoft: at least 4 above, 10–15 recommended all round
Under it: the deck, lid opens to 121

High shelf, front view: as planned, or as LP overflow
186 long, 30 deep, 39 clear to the ceiling; the right-hand 50.5 keeps Deco B and the framed 7-inch



about 1 m of books, a lidded box, a print

As planned



≈ 380 LPs spine-out at about 3.3 mm each; a 31.4 sleeve overhangs the 30 cm shelf by 1.4

As LP overflow

PC shelf, from above
82 along the wall × 28 out, at 155.4; keyboard tray in a 7 cm housing under it



side wall along the top · alcove to the right

Keyboard tray: 300 mm runners, rolls 25 cm out; the housing stops 12 short of the sofa end so the cables drop past it. Full-size wireless keyboard and mouse.

If the collection outgrows the wall
two overflow options, both face-out or spine-out to match

1 · LPs spine-out on the high shelf: about 380 more
No building; books move to the second option's top.

2 · A second record unit, west of the sofa, under the window
Bay 3 again (60 × 46 × 73, two face-out trays): 180 LPs, 240 singles.
At x 300–360 it leaves 123 cm to the table set for six (100 to walk).



front view, 60 × 73 (shown at half height)

All to one scale, 2.2 px per cm. Sizes: KUGGIS box 26 × 35 × 15 (IKEA); jewel case 142 × 125 × 10 mm (Wikipedia: Optical disc packaging); consoles, sleeves, Series X and the trays from the fit proof (section 3 of the plan).

The console shelves

1. Bay 1, from the top: rows of 16 (Xbox One), 16 (Xbox 360), then the game drawer's 25.2. Put the shelves on the pin holes nearest those heights.
2. Bay 2: the PS3's row of 23.7 at the top, on a full-width shelf. Below it, the PS2 and the PS1 sit beside the sub (next part).

Bay 2's sub column

3. Stand the 12 mm divider on the bay's floor, 23.4 cm from the bay's left inner face, tight under the PS3 shelf. Check it's plumb.

4. Fix it with one small angle bracket at the bottom (into the frame's bottom panel) and one at the top (into the PS3 shelf's underside).
5. Drill two 5 mm pin holes in the divider's right face at the height of the frame's pin holes for the PS2/PS1 shelf. Fit the part shelf on four pins.
6. Pair the sub with the soundbar now: its PAIR button is on the back, and it's hard to reach once it's in.
7. Lay the isolation riser on the column's floor, its front edge flush with the frame's front.
8. Stand the sub on it with the grille facing the room, centred: 1 cm to the divider, 1 cm to the frame's side. Slide it forward until its front is flush with the face frame's front (2 cm proud of the frame). Run its lead out of the back.
9. Check it touches nothing but the riser. If it does, a sheet of felt between, or nudge it.

The game drawer (bay 1, bottom row)

10. Fit the 350 mm runners to the frame's sides, level, so the drawer's base will sit about 1 cm above the frame's bottom panel. Drill pilot holes: BESTÅ sides are particleboard. If a screw won't hold, glue a 12 mm ply packer behind the runner.
11. Rout a 6 mm deep groove 1 cm up from the bottom on the inner faces of the sides, back and front panel, 9 mm wide, for the base.
12. Glue and screw the box together round the base: outside 54.3 × 35. Check the width against the runner maker's figure (usually the opening less 25.4 mm) before the glue sets.
13. Fit the cross divider 13 cm back from the front, and the two bin walls 16 cm in from each side. Glue in the bin floors with a slight back-rake, so the cases lean about 13°.
14. Felt the base and make the follower blocks.
15. Fit the runners' inner members to the box, hang it, and check it runs true empty.

The record trays (bay 3)

16. Screw the 400 mm runners to bay 3's sides with their bottom edges on the lines at 2.3 and 37.0 cm. Use every hole: a full tray is about 32 kg.
17. For each tray, glue and pin the two 40 × 8 sides and two 52.1 × 8 ends on top of the 53.9 × 40 base, round its edge.
18. Fit the 6 mm oak divider 32.6 cm from the left side's inner face: the LP trough on the left, the singles trough on the right.
19. Glue a raked rest (an oak offcut) at the back of each trough, so the sleeves lean back about 10°. Felt the troughs.
20. Fit the runners' inner members, hang the tray, and run it in and out empty, then with ten records.

The hidden plinth drawer

21. Make the box: the two 34 × 4.4 sides, the two 49.6 × 4.4 ends between them, the 6 mm base screwed underneath. It's 52 × 34 × 5.
22. Pack the plinth's inner faces under bay 2 with ply strips so the 300 mm runners sit the box's width plus their allowance apart, and fit the runners with the box 1.6 cm off the floor.
23. Felt the runners: the drawer sits under the sub.

The face frame and the drawer fronts

24. Cut the stiles (about 5 wide) and the rails from 2 cm oak. The bottom rail covers the plinth, 9 tall under bays 1–2, with an opening 54 × 6 for the plinth drawer, 1.6 up, under bay 2.
25. Scribe the two end stiles to the walls, then glue and pin the frame to the frames' fronts and bay 3. Notch the bottom rail's left end for the two socket leads.
26. Screw the oak lips onto the game drawer, the two trays and the plinth drawer from inside each box, with a 2–3 mm gap all round to the face frame.

27. Cut each finger pull underneath its lip: a cove routed along the back of the bottom edge.

Check before moving on: Every drawer runs smoothly, and the two trays clear each other by 2 cm.

Step 9 · Uprights and shelves · You + a helper · 2–3 days

Parts

- Two oak uprights, 165 × 30 × 4.5 (cut to fit)
- High shelf: 128.9 × 30 × 4 and 51.7 × 30 × 4
- Record ledge 128.9 × 7 × 2.4 and its lip
- Series X shelf 55 × 30 × 4
- Two oak cleats for the reveal, 30 long
- Oak wedges (2 + spares), 10 mm oak dowel for pegs and the uprights' dowels

Hardware and fixings

- Four concealed upright brackets
- 5 × 60 mm screws and plugs
- Wood glue
- LED strip in its aluminium channel (from the lighting phase)

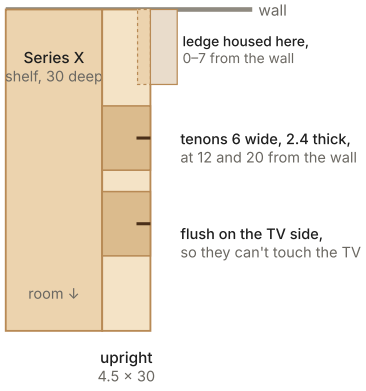
Tools

- Router with a straight bit and a guide batten
- Chisels 6, 12 and 25 mm, and a mallet
- Japanese pull saw
- Drill with a 10 mm brad-point bit and a 20 mm bit
- Dowel jig or dowel centres
- Combination square, marking knife
- Spirit level and laser
- Clamps
- A helper

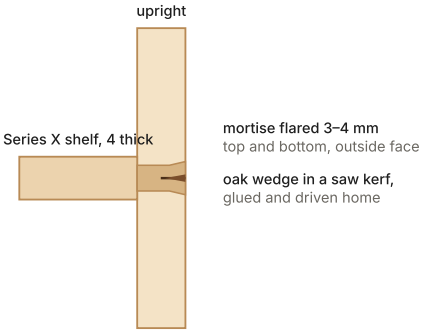
Artisan oak: how the joints work

Drawn to scale at 7 px per cm. The joints carry the load and are left on show, finished in hardwax oil

A · Wedged through-tenons, from above

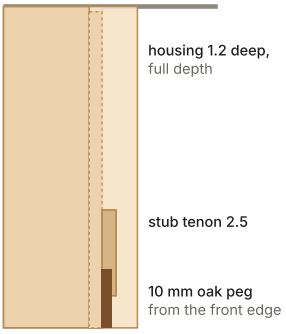


B · The same joint from the front



The kerf runs flat, so the wedge spreads the tenon up and down, along the upright's grain, and can't split it.

C · Housed shelf, pegged



high shelf / ledge
seen from above

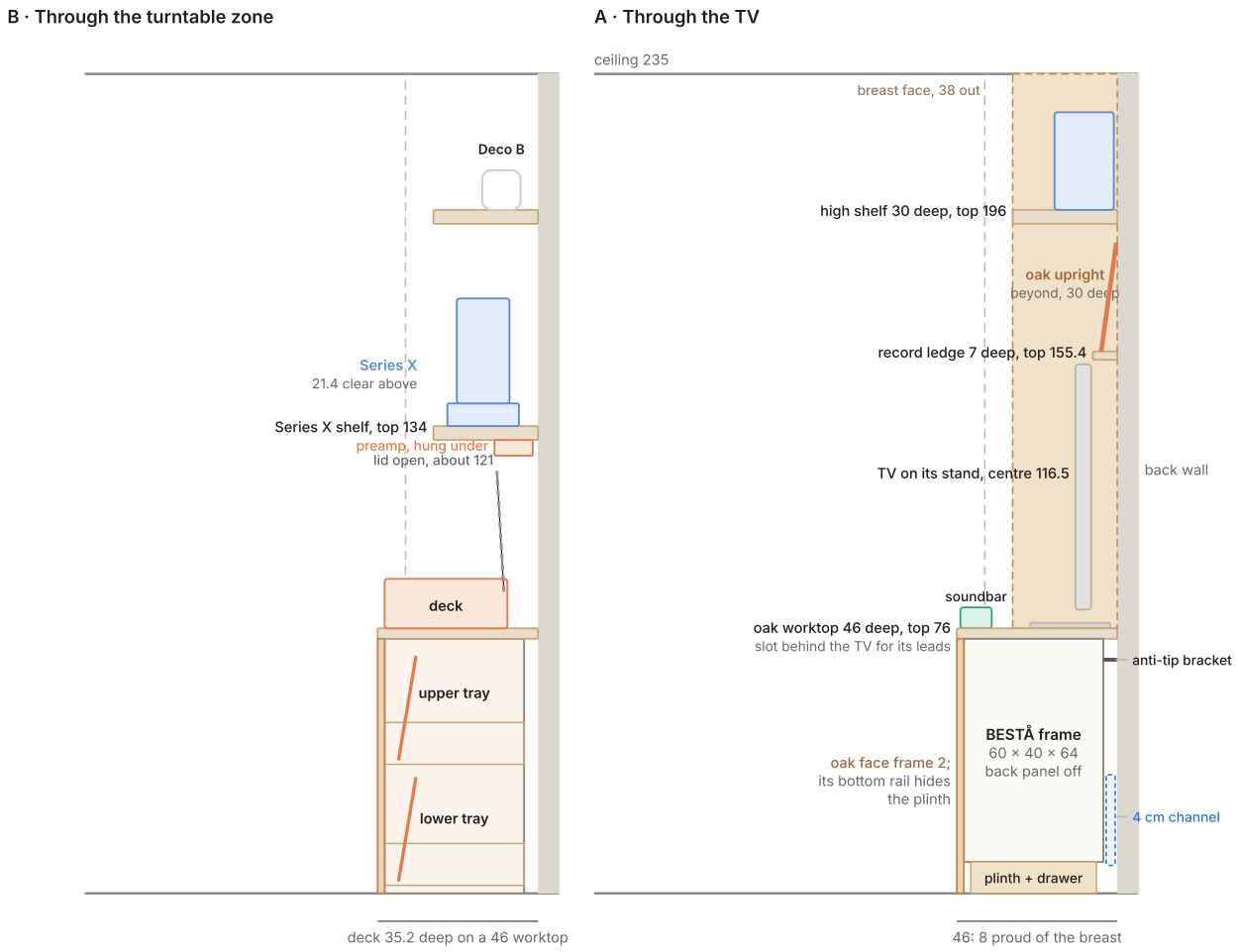
• Through-tenons (A, B): the Series X shelf (top 134) passes through the right upright on two tenons, 12 and 20 cm from the wall, wedged and planed flush: proud, they'd touch the TV

• Housed and pegged (C): the three lengths of the high shelf (top 196) and the record ledge (top 155.4). The peg shows as a dark dot on the upright's face.

• Uprights: 159 long, worktop to ceiling, each on two 10 mm dowels into the worktop, scribed to the ceiling, with two concealed brackets. The left one also screws to the side wall, and

How the wall goes together, layer by layer

Side sections to scale, room on the left. Dashed: the upright and the breast's face beyond the cut, and the deck's lid open



Cut the uprights to length

1. Measure from the worktop to the ceiling at each upright's position. The plan says 159; ceilings vary.
2. Cut each upright 2 mm under its measurement, then scribe its top to the ceiling as you did the worktop.
3. Mark each upright's faces: which side faces the TV, and the back edge that goes against the wall.

The housings (both uprights)

4. Measure from each upright's foot (the end on the worktop). The high shelf's housing is 116–120 up, the full 30 depth, 1.2 deep, on the TV-side face of both uprights.
5. The record ledge's housing is 77–79.4 up, 0–7 from the back edge, 1.2 deep, on the TV-side face of both.
6. On the right upright's other face (the turntable side), a second high shelf housing at 116–120, full depth, for the short length.
7. Clamp a straight batten as a guide, set the router 4 mm deep and rout, then step down to 12 mm in passes. Square the ends with a chisel. Practise on an offcut first.

The through-mortises (right upright)

8. Two mortises, each 6 wide and 2.4 tall, at 54.8–57.2 up from the foot, 12–18 and 20–26 cm from the back edge.
9. Mark them on both faces with a square and a knife. Drill most of each out with the 20 mm bit, halfway from each face, then chisel to the lines.
10. On the TV-side face, flare the top and bottom of each mortise outwards 3–4 mm, for the wedges.

The tenons (Series X shelf)

11. On the shelf's west end (the end that goes through the upright), mark two tenons 6 wide and 2.4 thick, centred in its 4 cm thickness, 12–18 and 20–26 from its back edge, 4.5 long.
12. Saw the shoulders and cheeks, and cut away the rest of the end back to the shoulder line.
13. Saw a kerf 3 cm deep across the middle of each tenon, parallel to the shelf's faces.
14. Test-fit through the mortises: snug, pushed in by hand.

The grooves

15. On each upright's TV-side face, rout a groove for the LED channel (usually about 1.7 wide × 0.8 deep: check yours), 2 cm in from the front edge, from 2 cm above the worktop to just under the high shelf's housing.
16. On the right upright's back edge, rout a 2 × 1.5 cable groove the full length, for Deco B's 12 V and patch leads, the Series X's HDMI and power, and the vinyl lead.
17. Notch both uprights' tops for the ceiling LED strip.

Dry-fit, then fix

18. Lay each upright on the floor with its shelves in their housings. Check every joint closes before any glue.
19. Drill two 10 mm holes 3 cm deep in each upright's foot, and matching holes in the worktop at the marked positions (dowel centres make this exact). Glue the dowels into the uprights only, so they can come out.
20. Lay Deco B's leads and the other cables into the right upright's back groove.
21. Stand the left upright on its dowels against the side wall. Screw it to the side wall through three countersunk holes, with plugs, and plug the holes with oak.
22. Stand the right upright on its dowels, check it's plumb with the laser, and fix it to the back wall with the concealed brackets.
23. Slide the long high shelf and the record ledge into their housings from the front.
24. Screw a 30 cm cleat to the breast's side face (the reveal) for the short high shelf, and another at the Series X shelf's underside height. Slide the short high shelf in from the front onto its cleat and into its housing.
25. Feed the Series X shelf's tenons through the mortises from the turntable side, onto its cleat. Glue the wedges and drive them into the kerfs from the TV side. When dry, saw them flush and plane: proud, they'd touch the TV.
26. Pegs: drill 10 mm through each upright's face into the housed shelf ends, 4 cm deep, about 4 cm from the front edge. Glue in a peg and saw it flush.

Check before moving on: Uprights plumb, shelves level, and the TV slides in with 2 cm each side.

Step 10 · Side wall and PC shelf · You · 1 day

Parts

- Wrap ledge 134 × 7 × 2.4 and its lip
- PC shelf 82 × 28 × 4
- Two oak cleats, 2 cm
- Keyboard tray: oak housing 70 × 28 × 7 (sides 7 × 2), 12 mm ply tray 54 × 26 with an oak lip

Hardware and fixings

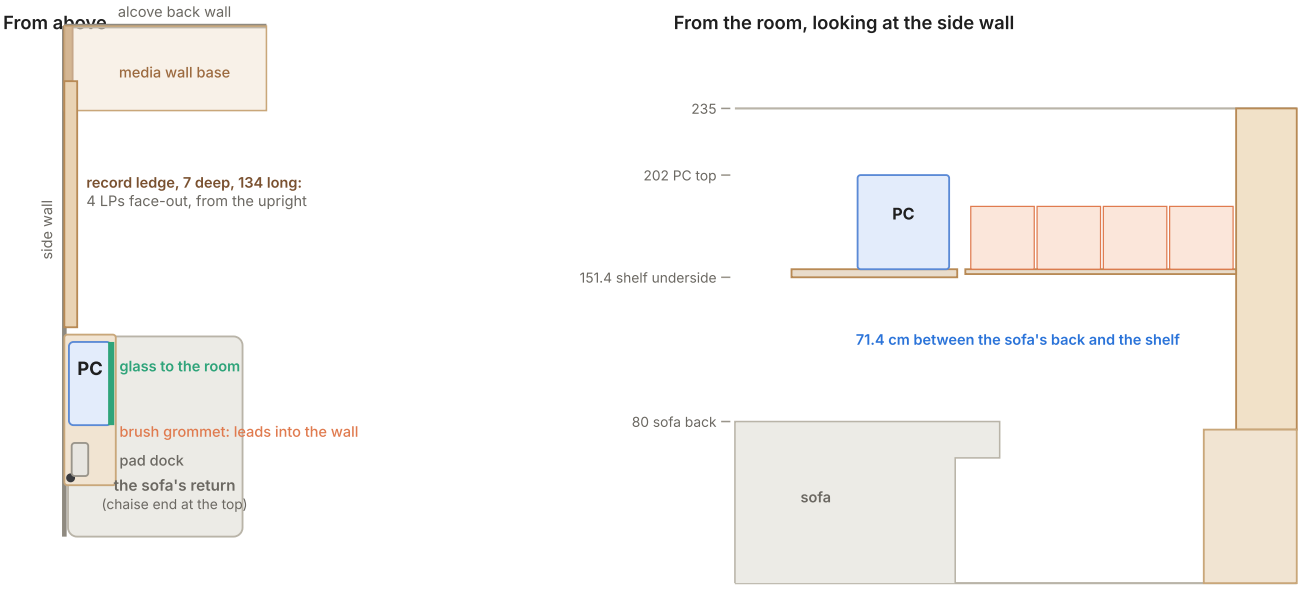
- Two concealed shelf brackets rated 40 kg+ a pair
- 300 mm light-duty runners, one pair
- An oak-coloured brush grommet
- 10 mm dowels (2)
- 5 × 60 mm screws and plugs

Tools

- SDS drill
- Drill-driver
- Dowel jig
- Spirit level and laser
- Jigsaw
- Clamps

The side wall: the ledge wraps the corner, the PC sits above the sofa

One oak line at 155.4 cm: the record ledge from the upright, then the PC shelf over the end of the sofa, clear of the TV's sightlines



The cleats

1. Screw the first cleat to the side wall with its top on the 153 line, from the left upright's front edge (30 cm out from the back wall) to 164 cm out. Plugs every 25 cm.
2. Screw the second with its top at 151.4 (the PC shelf's underside), from 168 to 250 cm out.

The wrap ledge

3. Drill two 10 mm dowel holes in the ledge's end and matching holes in the left upright's front edge, using the jig.
4. Glue the dowels, push the ledge onto them and down onto its cleat, and screw up through the cleat into it.
5. Glue the lip along its front edge.

The PC shelf and its tray

6. Fit the concealed brackets into the wall at the shelf's height, through the cleat, to the maker's instructions.
7. Cut the grommet hole in the shelf's back corner, over the top of the chased conduit.
8. Screw the keyboard tray's housing to the shelf's underside, stopping 12 cm short of the sofa end so the cables drop past it. Fit the runners inside it and slide the tray in.
9. Slide the shelf onto its brackets and down onto the cleat. Lean on it: it shouldn't move.

The PC's leads

10. Fit the brush grommet. Pull the HDMI (active optical: the end marked "source" at the PC), the Cat6 and the power lead down the conduit with the draw wire.
11. At the bottom, they run behind the skirting into the plinth's notch and the channel.

Check before moving on: One level line with the ledge over the TV, and the PC shelf holds your weight leaning on it.

Step 11 · The dining wall · You + a helper · 1 day

Buy first: 6 Dining wall (the buy list).

Parts

- Solid oak: table centre 101 × 80 × 2.7 (edge-glued staves), two breadboard ends 80 × 7 × 2.7, the leaf 60 × 80 × 2.7, the backboard 110 × 22 × 3 (from step 7), two turn buttons 6 × 2.5 × 2.2
- Eight legs: tapered oak 71 × 4.5 to 3 on M8 hanger bolts, or Emuca 710 mm steel legs

Hardware and fixings

- Two Häfele Hebgo 287.43.428 folding brackets
- Union PowerLoad 100 × 88 lift-off hinges, one 3-pack, all the same hand
- Eight brass M8 threaded inserts
- Two leaf-alignment dowels and two table clips
- A magnetic catch
- 5 × 70 mm screws and brown plugs
- Felt pads
- Hardwax oil, glue, 10 mm oak dowel for pegs

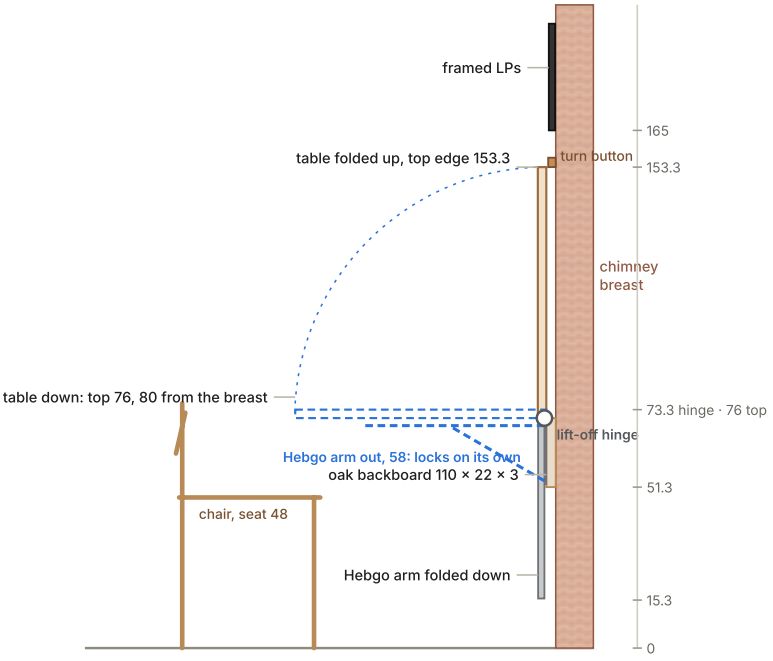
Tools

- Router with a straight bit and a guide
- Drill with a 10 mm brad-point bit, and the inserts' bit with a depth stop
- Jigsaw and sander
- Chisels
- SDS drill
- Spirit level
- Clamps
- A helper

The dining wall in section: folded, down, or off the wall

Through the chimney breast at the table's centre, to scale, room on the left. Solid: folded. Dashed blue: down for four

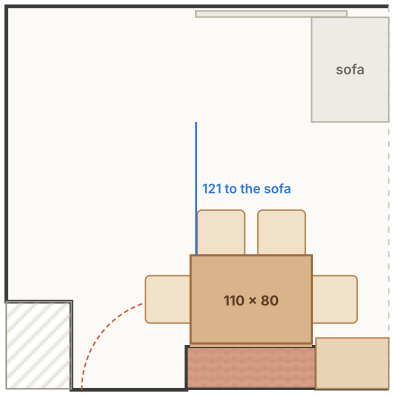
- Everyday: folded flat, about 7 cm off the breast, held at the top by two turn buttons.
- For four: lift the top edge a little, turn the buttons, lower it onto the two brackets as they swing out and lock.
- For six or eight: with it down, slide it 5 cm sideways off the lift-off hinges' pins, screw in four legs (710–730 mm),
- and stand it in the room: the top is then 73.7–75.7 cm. Two people lift it: about 18 kg.



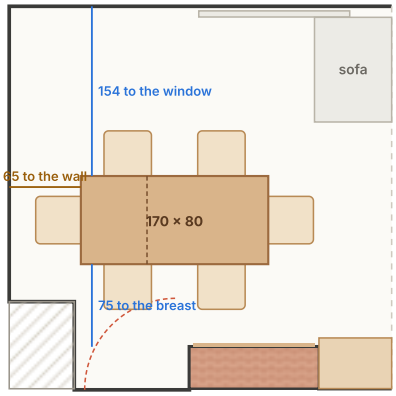
The dining table for four, six or eight

To scale, the west half of the room from above, window at the top. Red: the door's swing. Blue: the clearances that matter

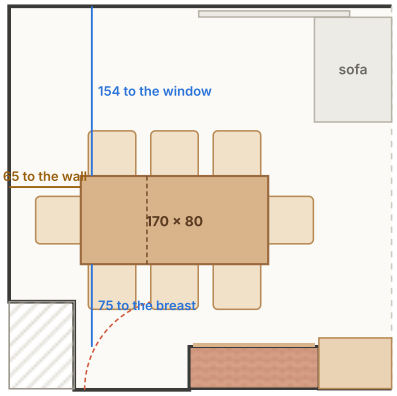
Four, at the wall
everyday: the table down on its brackets



Six, in the room
lifted off, with the leaf: 170 × 80 on eight legs



Eight, with the leaf
plus a 60 × 80 leaf on four more legs



- Per diner: 60 cm of table edge (55 at a squeeze). Table edge to a wall: 60 cm is a tight minimum, 75 to aim for, 100+ to walk behind someone seated.
- Six and eight: slide the coffee table to the window first. The dashed red arc is the door's swing if it opens into the room: the chairs on that side need it shut while you eat.
- Chairs drawn at IKEA TERJE size (43 × 49, seat 48): 28 cm from seat to table top, the usual 25–30.

Order the oak

1. Ask the timber merchant for the centre panel as edge-glued staves, 101 × 80 × 2.7: 96 between the shoulders plus a 2.5 tongue at each end. Most will also cut the breadboard ends and the leaf, and radius the corners.

The breadboard ends

2. On each end of the centre panel, rout a tongue 2.5 long and 0.9 thick, centred in its thickness: take 0.9 off each face with the router against a batten.
3. In each breadboard end's inner edge, rout a groove 0.9 wide and 2.6 deep, centred, full length.
4. Dry-fit: the breadboard ends flush with the panel's long edges, the top flat across the joint.
5. Drill three 10 mm peg holes down through each breadboard end and the tongue in it: one at the centre and one 8 cm in from each long edge, 1.3 cm from the breadboard's inner edge.
6. Take the ends off. Elongate the two outer holes in each tongue to 16 mm long, along the panel's width: the panel moves with the seasons, the breadboard doesn't.
7. Glue only the middle 10 cm of each tongue and groove, clamp the ends on, and drive the pegs. Glue the outer pegs into the breadboard only, not the tongue. Saw flush and sand.

Shape and finish

8. Radius all four corners of the top to 5 cm, and the leaf's two outer corners: jigsaw outside the line, then sand to it.
9. Chamfer the underside edges 1 cm, so the top reads slim.
10. Sand both to 180. Oil both faces equally, every edge too: a top oiled on one side cups.

Inserts and legs

11. Mark the leg positions on the underside: centred in each breadboard end's 7 cm width, 6 cm in from the long edges. On the leaf, 6 cm in from each edge at its four corners.
12. Drill each insert's hole to the maker's size with a depth stop, never deeper than 1.8 cm: the top is only 2.7 thick.
13. Wind the inserts in with a bolt and a locked nut, square to the surface.

14. Screw in all eight legs and stand the table and leaf on the floor: the top at 73.7 on 710 legs. Check it's level, then take the legs out and store them with the leaf.

The backboard

15. Hold the backboard on its marks: centred on the breast, top at 73.3, level.
16. Drill through it and into the breast every 20 cm in two rows, 3 cm from its top and bottom edges. Plug, then fix with 5 × 70 countersunk screws.

The brackets

17. Screw the two Hebgo brackets to the backboard's face, 80 cm apart centre to centre (15 cm in from each end), their tops level with its top edge.
18. Use long screws through the backboard into plugs in the breast, as many as the bracket's holes take: the 500 kg rating depends on the fixing.
19. Raise both arms until they click and lock. Lay a level across them.

Hang the table

20. Screw the three lift-off hinges along the backboard's top edge, 10 cm from each end and one in the middle, pin halves on the backboard.
21. Screw the other halves to the underside of the table's back edge (the long edge that meets the wall) at the same spacing, all the same hand.
22. With your helper, lower the table onto the pins and let it rest on the arms. Its top should be at 76, level, in line with the alcove's worktop.

Folded

23. Lift it to vertical: its top edge is at 153.3. Screw the two oak turn buttons to the breast just above it, 15 cm in from each end.
24. Fit the magnetic catch behind the top edge. Fold the brackets flat.
25. Fold it down and up three times: it shouldn't bind, both arms should click, and it shouldn't move when you lean on it.

The leaf

26. Drill two 10 mm holes in the leaf's edge and matching holes in the table's west breadboard end, and glue the alignment dowels into the leaf.
27. Screw the two table clips under the joint so they pull the leaf to the table.

Check before moving on: It folds up and down without binding, both brackets click and lock, and it doesn't move when you lean on it.

Step 12 · Lighting · You · ½ day

Buy first: 7 Lighting (the buy list).

Parts

- 2700K LED strip, two aluminium channels and a driver
- 6500K bias light for the TV
- Eight warm-white dimmable E14 bulbs, 6 W max
- A lamp and a 2700K bulb

Tools

- Small screwdriver
- Step ladder

1. LED strip into each upright's groove, wired to its driver on the fused spur.
2. The ceiling strip through its notches in the uprights; the bias light on the back of the TV.
3. Aim both spot bars (the table in the Paint and light tab says which head goes where).
4. The lamp by the sofa, with a 2700K bulb.

Check before moving on: From the sofa, no spot head shows in the screen or shines in your eyes.

Step 13 · Connect and finish · You · 1 day

Buy first: 8 TV, consoles and AV, 9 Vinyl kit, 10 Finishing (the buy list).

Parts

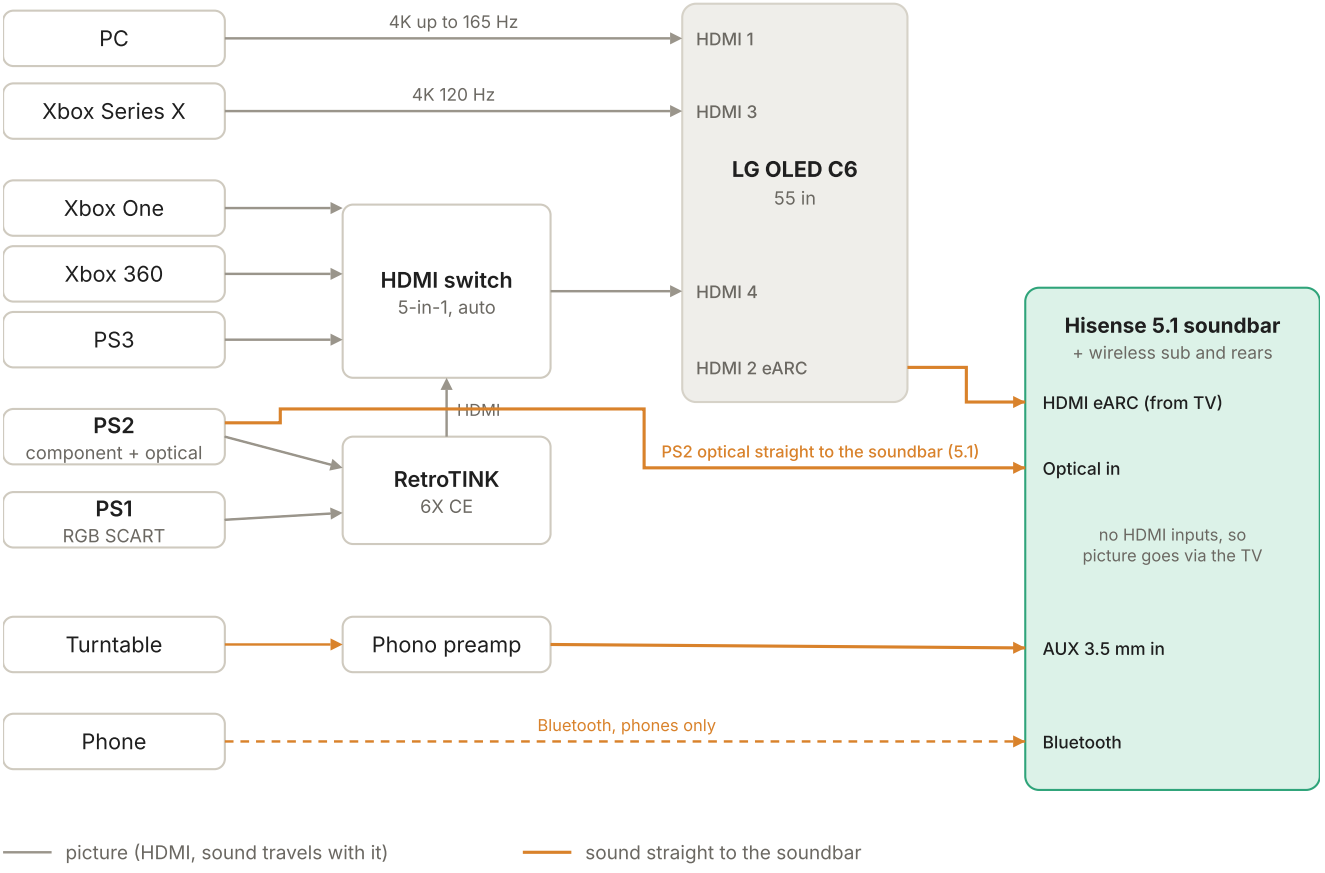
- HDMI switch, RetroTINK, HDMI leads, optical and SCART cables, the active optical HDMI for the PC
- Stylus, preamp, RCA-to-3.5 mm lead, record brush
- Labels

Tools

- Phone with the Deco app
- Velcro ties

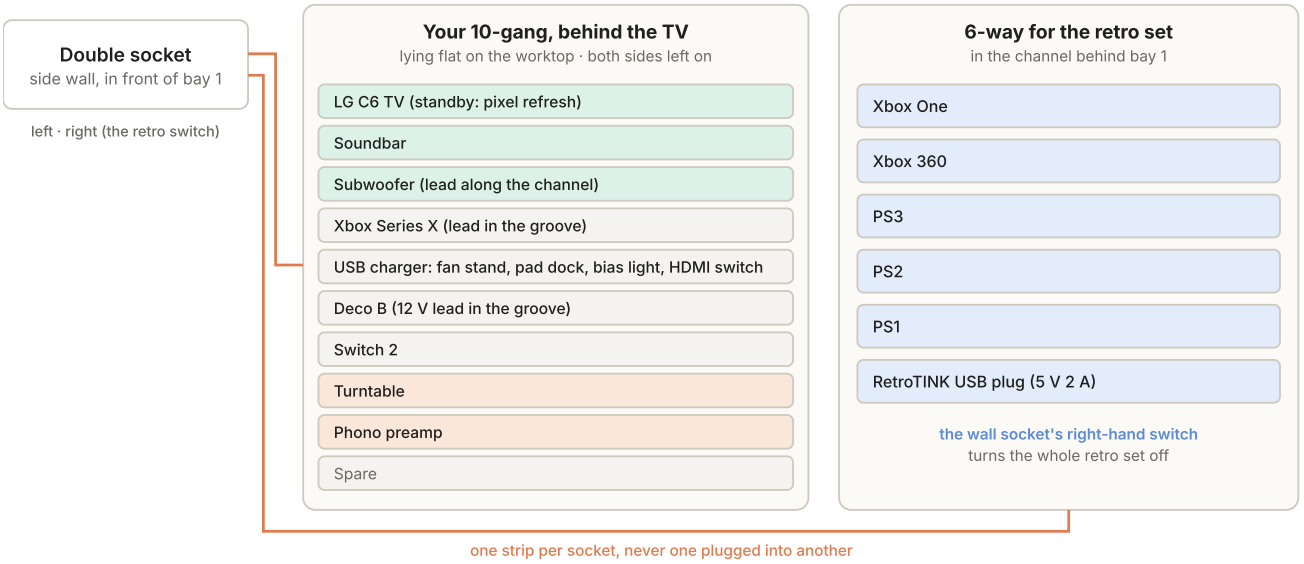
Everything sounds through the Hisense: directly where it can, via the TV where it can't

The soundbar has no HDMI inputs, so HDMI sources reach it over eARC; vinyl, PS2 audio and phones go straight in



Power map: your 10-gang plus a 6-way for the retro set

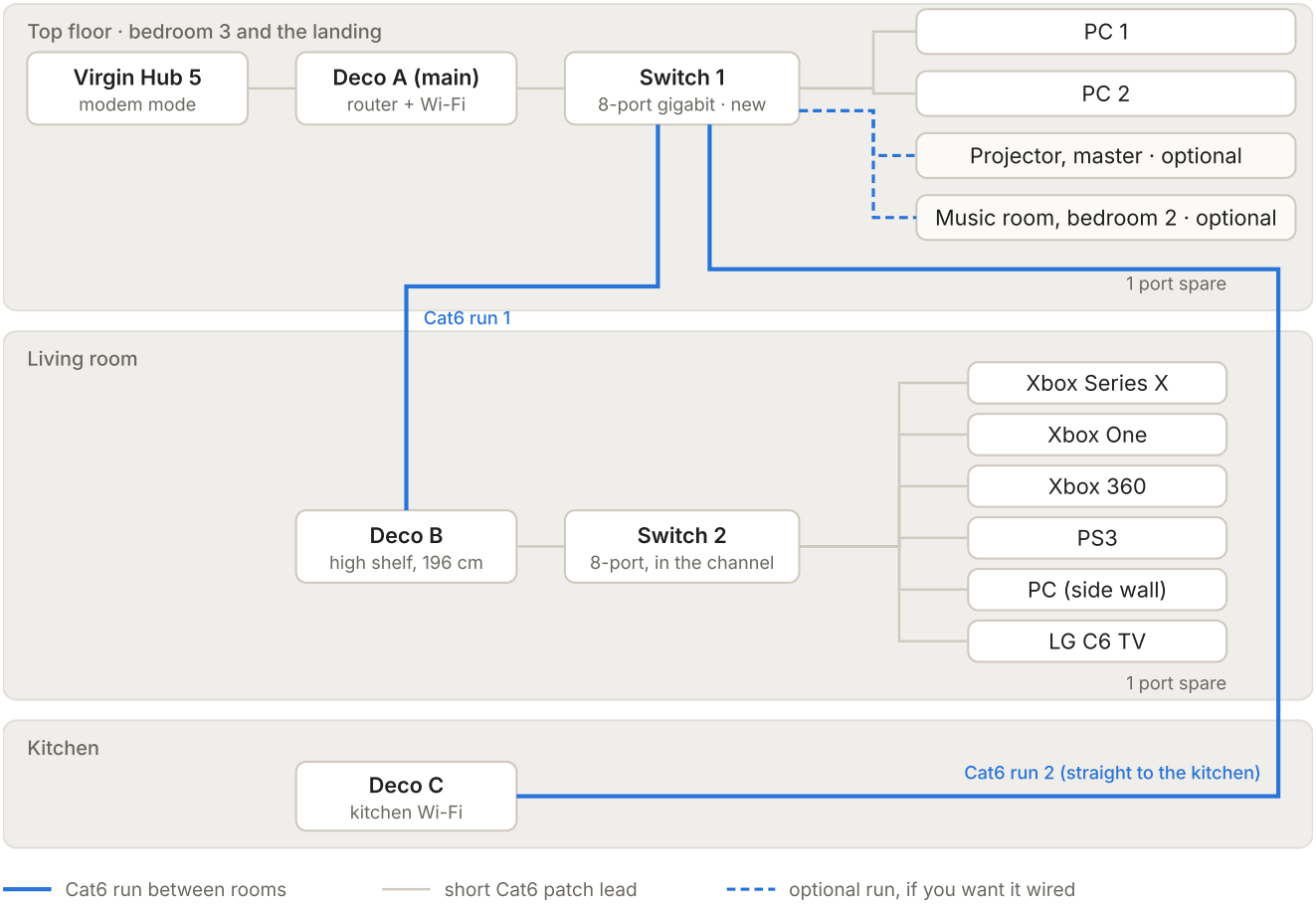
Everything modern on the left outlet, always on; the five retro consoles and the RetroTINK on the right, switched at the wall



- The socket on the side wall is a single today: the electrician swaps it for a double, at the same spot.
- The retro switch moves to the wall: with the TV 2 cm from the left upright, the 10-gang's own switches can't be reached any more.
- PC and rear speakers: a new double socket low on the side wall behind the sofa; the PC (500–850 W) stays off these strips.
- The sub plugs into the 10-gang: its lead drops through the worktop's slot to the back of bay 2. Both spot bars use existing ceiling points.
- Load check: the 10-gang about 0.7 kW with everything on, the 6-way up to about 0.8 kW with a launch PS3 running; each well under 13 A (3 kW).
- Check the 10-gang is surge-protected; if not, use a surge-protected one for the TV and consoles. Label both outlets.

Every floor has its own Cat6 link back to bedroom 3

Deco A and Switch 1 live with the Hub 5 and the gaming PCs in bedroom 3; two spare ports can wire the projector and the music room



1. TV inputs: HDMI 1 the PC, HDMI 2 the soundbar (eARC), HDMI 3 the Series X, HDMI 4 the HDMI switch (Xbox One, 360, PS3, RetroTINK).
2. The PS2 by component into the RetroTINK and by optical to the soundbar; the PS1 by RGB SCART.
3. The deck to the preamp, the preamp to the soundbar's AUX; set the tracking weight; soundbar in Music mode for vinyl.
4. Network stage 2: plug each Cat6 run into its Deco, check Deco B and C show "Ethernet", and give the Series X and PC high priority.
5. Label both outlets of the double socket and each HDMI switch input; set the sub and rear levels from the sofa.

Check before moving on: Everything on the network's "done when" list, and every source plays through the soundbar.

Shopping list

About £2,589–4,897 plus the electrician, in build order.

Status	Item	Qty	Estimate (£)	Paid (£)
1 NETWORK, NOW				
☐ To do	Put the Virgin Hub 5 into modem mode			
☐ To do	Set up Deco A, Switch 1 and the PCs in bedroom 3			
☐ To do	Add Deco B and Deco C over Wi-Fi for now; speed-test each room and the Xbox			
☐ To buy	Cat6 patch leads, 0.5–2 m	11	15–20	
☐ To buy	TP-Link TL-SG108 8-port switch: one for the living room, one to replace the 5-port in bedroom 3	2	36–50	
2 ELECTRICIAN AND CABLING				
☐ To do	Book one electrician visit: Cat6 from bedroom 3 down through the airing cupboard to the alcove's right corner (Deco B) and on to the kitchen (Deco C), a spare pulled with each, and the optional runs to the master and bedroom 2; swap the side-wall socket for a double **at the same spot**; a new double socket low on the side wall behind the sofa for the PC and the rear speakers **(new)**; LED-compatible dimmers for both ceiling points; a fused spur for the LED drivers; fit both spot bars, on the existing rose between the TV and the sofa and the existing point by the door **(changed: no new point)**; optionally a conduit to TV height for a future wall mount **(from v6.1)**		Quote	
☐ To buy	Cat6 solid pure copper, round (not CCA): two runs, four with the optional ones	2–4 runs	30–60	
☐ To buy	6-way surge-protected extension for the retro set (Xbox One, 360, PS3, PS2, PS1, RetroTINK) **(changed: switched at the wall)**	1	20–35	
☐ To buy	4-port USB charger, 5 V 2 A per port (HDMI switch, fan stand, pad dock, bias light)	1	10–20	
☐ To buy	USB plug, 5 V 2 A or more, for the RetroTINK on the retro 6-way	1	5–10	
☐ To buy	Cable kit: brush grommets, velcro ties, labels	1	20–30	
☐ To buy	Flat two-compartment trunking for the 4 cm channel behind the base, about 2 m	1	10–20	
☐ To do	Same electrician visit: chase a 25 mm conduit into the side wall for the PC's leads, from behind the PC shelf down to skirting height and behind the skirting to the base's plinth; make good and paint (the wall is solid) **(new: replaces the spine and trunking)**		Quote	
☐ To buy	25 mm oval conduit and a draw wire for the chased PC drop, about 4 m; one oak-coloured brush grommet for the PC shelf **(new)**	1	10–25	
☐ To buy	PC tidy kit: braided sleeve, right-angle C13 lead, shelf-notch grommet	1	10–20	
3 PAINT				
☐ To do	Sample Farrow & Ball Card Room Green No. 79 on an A4 board; check it by day and with the LED strip on			
☐ To buy	Card Room Green: Estate (or Modern) Emulsion for walls, ceiling, alcove and chimney breast; Estate Eggshell for skirting, door and frame; primer and undercoat		200–500	
☐ To buy	Woodwork primer, sanding pads and sugar soap for the white skirting and door	1	20–40	
☐ To do	Paint the whole room before anything is fixed: alcove and breast included; skirting and door from white to eggshell			
4 BASE UNIT				
☑ Done	Check the alcove back wall for fixings (solid)			

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To do	Measure the alcove's width at the floor, 76, 134, 155 and 196 and the ceiling, and the floor's level across it			
☑ Done	Check the side wall where the ledge and PC shelf go, and the chimney breast where the dining wall goes: solid (you confirmed) **(new)**			
☐ To buy	Two IKEA BESTÅ 60×40×64 frames, white stained oak effect, for bays 1–2	2	70–200	
☐ To buy	Plinth for bays 1–2: 18 mm ply ladder frame 120 × 38 × 9 cm with a clear bay under bay 2, plus shims	1	15–30	
☐ To buy	Hidden plinth drawer: oak front 54 × 6 with a finger-pull groove underneath, 12 mm ply box 52 × 34 × 5, 300 mm full-extension runners **(changed: no push-to-open at floor level, as v6.1 said)**	1	30–50	
☐ To buy	Bay 3 carcass for the records: 18 mm birch ply, 60 × 40 × 73 cm from the floor, connector bolts **(changed: holds the trays)**	1	45–80	
☐ To buy	Two face-out record tray kits: 12 and 9 mm birch ply cut to size, 6 mm oak dividers, 15 mm oak lips, raked rests, felt, two pairs of 400 mm runners rated 68–70 kg	2	150–240	
☐ To buy	BESTÅ shelves, 56 × 36 cm (20 kg max each), two each for bays 1 and 2	4	12–32	
☐ To buy	The sub in bay 2: an isolation riser for it (dense foam or sorbothane, about 22 × 28 × 3), 12 mm birch ply for the divider (35.3 × 38) and the part shelf (32.2 × 36), four shelf pins, felt pads **(new in v13.5: the sub moves into the media wall)**	1	25–45	
☐ To buy	Anti-tip brackets and 4 cm spacer blocks: both frames and bay 3	1 set	15–25	
☐ To do	Measure about ten games of each kind (PS2, Xbox 360, Xbox One) before cutting the game drawer			
☐ To buy	Game drawer kit: 12 mm birch ply cut to size, 350 mm full-extension soft-close runners (35–49 kg), oak lip, follower block, felt	1	50–90	
☐ To do	Build the plinth and bay 3, fix everything to the wall, fit the shelves, drawers and trays; place the TV and soundbar			
5 OAK JOINERY				
☐ To buy	Two solid oak uprights, 165 × 30 × 4.5 cm (cut to 159), quarter-sawn if possible	2	150–300	
☐ To buy	Solid oak for the shelves: high shelf in two lengths (128.9 and 51.7 × 30 × 4), Series X shelf 55 × 30 × 4, record ledge 128.9 × 7 × 2.4, wrap ledge 134 × 7 × 2.4, PC shelf 82 × 28 × 4, lips **(changed)**	1 set	200–340	
☐ To buy	Oak wedges, 10 mm oak dowel for pegs and dowels, concealed upright brackets; two concealed shelf brackets for the PC shelf, rated 40 kg+ a pair	1	30–60	
☐ To buy	One solid oak worktop, 300 × 62 cm, full stave: the alcove top (186 × 46) and the dining wall's backboard (110 × 22) from the offcut, chamfered fronts **(changed)**	1	250–450	
☐ To buy	Oak face frame, 20 mm, 186 cm for the base	1	60–120	
☐ To buy	Keyboard tray for the PC shelf: 7 cm oak housing under it, 12 mm ply tray 54 × 26 with an oak lip, 300 mm light-duty runners **(new: back from v6.1)**	1	25–45	
☐ To do	Cut the joints and fit the uprights, shelves, ledges and face frame; notch the uprights for the ceiling strip			
6 DINING WALL				
☐ To buy	Solid oak table top 110 × 80 × 2.7 with pegged breadboard ends, and a 60 × 80 × 2.7 leaf (edge-glued staves from a timber merchant, who can also radius the corners) **(new)**	1 set	220–400	
☐ To buy	Two Häfele Hebgo heavy-duty folding brackets, 287.43.428 (580 arm, 500 kg a pair) **(new)**	2	70–120	

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To buy	Union PowerLoad lift-off hinges, 100 × 88, one 3-pack (all the same hand) **(new)**	1	16–20	
☐ To buy	Eight legs on screw-in plates: tapered oak legs on M8 hanger bolts (to match), or Emuca Ø60 × 710 mm steel legs (cheaper); four for the top, four for the leaf **(new)**	8	70–160	
☐ To buy	Two oak turn buttons, two leaf-alignment dowels and two table clips for the leaf, felt pads **(new)**	1	10–20	
☐ To buy	Four folding chairs, IKEA TERJE or similar (skip if yours fit: seat about 45–48) **(new)**	4	100–160	
☐ To buy	Two LP frames for the chimney breast, hung from 165 cm	2	20–40	
☐ To buy	Eight M8 brass threaded inserts for the legs, four in the top's breadboard ends and four in the leaf, set into the underside instead of plates **(changed in v13.5: the leaf's four)**	8	12–20	
☐ To do	Fix the backboard to the breast, hang the brackets and the top, check it folds and locks; set the legs to the right height			
7 LIGHTING				
☐ To buy	LED for the upright grooves: 2700K strip, two aluminium channels, driver; 6500K bias light for the TV	1	40–90	
☐ To buy	Two Dunelm Elements Wolston 4-light spotlight bars (£49 each), plus eight warm-white (2700K) dimmable E14 LED bulbs, 6 W max **(changed: two)**	2	120–160	
☐ To buy	A floor or table lamp by the sofa with a 2700K warm LED bulb: the low layer a green room needs **(new: from v6.1)**	1	30–80	
☐ To buy	Later: aluminium channel and opal diffuser for the existing ceiling LED strip (optional)	1	0–80	
8 TV, CONSOLES AND AV				
☐ To buy	5-in-1 4K60 HDMI switch with remote	1	20–35	
☐ To do	Measure the PC-to-TV cable route (about 4–5 m along the skirting)			
☐ To buy	Active optical HDMI cable for the PC, certified Ultra High Speed (48 Gbps), about 5 m	1	20–60	
☐ To buy	High Speed HDMI leads, 1–2 m; two 90° HDMI adapters in case the TV's plugs need them	5 + 2	25–40	
☐ To buy	RetroTINK-6X CE upscaler for the PS2 and PS1 (order on restock; UK price includes VAT and import fees)	1	200–240	
☐ To buy	PS2 component video cable, good quality	1	10–20	
☐ To buy	Optical (TOSLINK) cable, 2 m, PS2 straight to the soundbar	1	5–10	
☐ To buy	RGB SCART cable for the PS1	1	8–15	
☐ To do	Connect the TV inputs; set up the PS2 and RetroTINK			
9 VINYL KIT				
☐ To buy	Replacement stylus for the turntable (exact match for its model)	1	15–40	
☐ To buy	External phono preamp (only if the deck has a PHONO/LINE switch)	1	40–100	
☐ To buy	RCA-to-3.5 mm lead, 2 m: preamp under the Series X shelf to the soundbar's AUX **(changed: shorter)**	1	5–10	
☐ To buy	Isolation feet for the turntable; a carbon-fibre record brush	1	20–40	
☐ To do	Set the tracking weight; soundbar in Music mode			
10 FINISHING				
☐ To do	Turn on HDMI-CEC (SIMPLINK); PS3 Control for HDMI off			

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To do	Deco app: prioritise the Series X and PC, guest Wi-Fi, automatic firmware			
☐ To do	Label both outlets of the double socket (left: everything, right: retro) and each HDMI switch input			
☐ To do	Set the sub and rear speaker levels from the sofa			